Sticking the Landing for Stochastic Gaussian Natural-Gradient Variational Inference: Global Theory and a Mean-Block Variance-Reduction Study

Abstract

This report is a self-contained account of the Sticking-the-Landing (STL) trick for stochastic Gaussian natural-gradient variational inference. It combines the global convergence *theory* (Part I) with a numerical *variance-reduction study* (Part II) of the same schemes, so that the empirical findings can be read directly against the estimates that predict them.

Part I sharpens the global analysis along three axes. A *covariance bootstrap* replaces the frozen lower covariance bound $\min(\lambda_{0,\min}, 1/\beta)$ used in the existing global proofs by the exact lower envelope of the covariance recursion, improving the warm-up dependence on the initial covariance from inverse, $1/\lambda_{0,\min}$, to logarithmic, $\log_+(1/\lambda_{0,\min})$, and reorganizing the iteration complexity into three transparent stages (covariance burn-in, global localization, local convergence). A short *Wasserstein transport bootstrap* followed by a permanent switch to Fisher–Rao removes even the logarithmic burn-in in the deterministic setting: a single Bures–Wasserstein forward–backward step of size c/β creates the floor $C \succeq c\beta^{-1}I$. On the stochastic side, an *intrinsic STL variance identity* shows that the STL noise is bounded by the squared Fisher–Rao gradient norm plus a multiple of the intrinsic Hessian fluctuation $\Psi(a) = \mathbb{E}_{X\sim\rho_a}\|C^{1/2}(\nabla^2 V(X) - A(a))C^{1/2}\|_{\mathrm{F}}^2$. Since $\Psi \equiv 0$ for a Gaussian target after whitening, the single-sample STL schemes have *no* asymptotic stochastic floor for Gaussian targets, and a floor controlled by the condition number $\kappa = \beta/\alpha$ (rather than by raw powers of $\beta, \alpha, \lambda_{\max}$) for general strongly log-concave targets.

Part II isolates the effect of mean-block STL numerically. It replaces the plain posterior-score mean estimator $b_{\mathrm{base}}(\theta) = \mathrm{score}_{\mathrm{post}}(\theta)$ by the score-*residual* estimator $b_{\mathrm{stl}}(\theta) = \mathrm{score}_{\mathrm{post}}(\theta) + C^{-1}(\theta - m)$, which is exactly the STL score estimator of the theory (Proposition on pointwise vanishing mirrors Theorem 11.2). Two experiments – an estimator-level variance comparison at fixed states and an algorithm-level trajectory comparison – on a well-specified anisotropic Gaussian target and a misspecified smooth $\log\cosh$ target confirm the theory: for the Gaussian target the STL mean estimator has *exactly zero* variance whenever $C = C_\star$ and the algorithm's stochastic noise floor collapses to machine precision (the $\Psi \equiv 0$, no-floor prediction); for the $\log\cosh$ target STL reduces the floor but cannot remove it, and under covariance under-dispersion it can even *increase* the variance. Mean-STL is thus an essentially cost-free, practically effective variance-reduction device for correctly specified Gaussian families near the optimum – but, consistent with the theory, not a universal remedy.

Contents

Part I
Theoretical framework: covariance bootstrap, transport warm-start, and global stochastic STL bounds

1 Purpose of the note

This part records two refinements of the convergence analysis for the Gaussian natural gradient flow and its two deterministic discretizations (the Riemannian/geodesic scheme and the KL/Bregman scheme), together with their single-sample stochastic counterparts.

Deterministic refinement: covariance bootstrap. The previous global proofs use the fixed lower covariance bound

$$C_t \succeq \lambda_{\min} I, \qquad \lambda_{\min} = \min\left\{\lambda_{0,\min}, \tfrac{1}{\beta}\right\},$$

for the entire global phase. This gives a valid but pessimistic warm-up time proportional to $1/\lambda_{0,\min}$. When the covariance is too small it grows; tracking this growth through a covariance bootstrap improves the dependence on the initial lower covariance from inverse to logarithmic,

$$\frac{1}{\lambda_{0,\min}} \quad \rightsquigarrow \quad \log_+ \frac{1}{\lambda_{0,\min}}.$$

After this refinement, the iteration complexity is a three-stage phenomenon:

$$\boxed{\text{covariance burn-in} \;+\; \text{global localization} \;+\; \text{local convergence.}}$$

Transport-bootstrap refinement: Wasserstein warm-start. The logarithmic covariance burn-in is intrinsic to pure Fisher–Rao dynamics when the initial covariance is severely underdispersed. However, it can be removed algorithmically by using Wasserstein transport only as a short warm-start. The reason is that, in the small-covariance regime, Fisher–Rao covariance growth is multiplicative, whereas Bures–Wasserstein covariance growth is additive and scale-independent. We therefore add a standalone deterministic section proving a two-stage transport-bootstrap natural-gradient principle: first lift the covariance to a fixed curvature-scale floor, then switch permanently to Fisher–Rao for global localization and local convergence. This avoids tuning an adaptive Wasserstein–Fisher–Rao schedule and avoids paying the cost of a transport step at every iteration.

Stochastic refinement: intrinsic STL variance. The same covariance bootstrap holds *pathwise* for the stochastic schemes, because the sampled Hessian still obeys the almost-sure spectral bound $\alpha I \preceq \nabla^2 V(X_n) \preceq \beta I$. The remaining obstacle is the stochastic variance term. The existing global stochastic bounds use raw Euclidean/Frobenius variance estimates, which propagate large powers of β, α, and $\lambda_{\max}$. We instead analyze the single-sample sticking-the-landing (STL) estimator in the intrinsic Fisher–Rao norm. The key quantity is the intrinsic Hessian fluctuation

$$\Psi(a) := \mathbb{E}_{X\sim\rho_a}\left[\left\|C^{1/2}\big(\nabla^2 V(X) - A(a)\big)C^{1/2}\right\|_{\mathrm{F}}^2\right], \qquad A(a) := \mathbb{E}_{\rho_a}[\nabla^2 V(X)].$$

We show that the intrinsic STL noise is bounded by the squared Fisher–Rao gradient norm plus a multiple of $\Psi(a)$. Since $\Psi \equiv 0$ for a Gaussian target after whitening, the single-sample STL schemes then have no asymptotic noise floor for Gaussian targets, and a κ-controlled floor for general strongly log-concave targets. Finally, under Assumption 2.2, we add the missing first-stage stochastic estimate: during covariance burn-in, the energy gap is controlled in expectation with an explicit accumulated error proportional to $\Delta t^2 N_{\mathrm{cov}} \beta N_\theta^2 L_{H,\mathrm{g}}^2 \lambda_{\max}^4$. Thus the stochastic theory becomes a closed three-stage statement rather than a post-burn-in statement with the quantity $\mathbb{E}\Delta_{N_{\mathrm{cov}}}$ left uncontrolled.

Status of the constants. All constants below are explicit. The deterministic constants and the Gaussian-target and Riemannian-STL stochastic constants are sharp up to the stated elementary inequalities. The constants in the general KL-STL stochastic floor are honest but *not optimized*: the covariance perturbation of the KL scheme is intrinsically a matrix-inverse perturbation, and the explicit bound we give carries extra factors of κ that we do not attempt to remove. We flag this explicitly where it occurs (Remark 12.9). The qualitative conclusions—logarithmic covariance dependence, three-stage complexity, vanishing stochastic floor for Gaussian targets, and condition-number-only floor for general targets—do not depend on optimizing these constants.

2 Setup and notation

We consider the Gaussian variational family

$$\mathcal{P}_{\mathrm{G}} := \left\{\rho_a = \mathcal{N}(m, C) : a = (m, C) \in \mathcal{A} := \mathbb{R}^{N_\theta} \times \mathbb{S}_{++}(N_\theta)\right\},$$

and the objective

$$\mathcal{E}(a) := \mathrm{KL}(\rho_a \,\|\, \rho_{\mathrm{post}}), \qquad \rho_{\mathrm{post}}(\theta) \propto \exp(-V(\theta)), \qquad a_\star = (m_\star, C_\star) \in \operatorname*{argmin}_{a\in\mathcal{A}} \mathcal{E}(a).$$

We write the energy gap as

$$\Delta(a) := \mathcal{E}(a) - \mathcal{E}(a_\star), \qquad \Delta_0 := \Delta(a_0), \qquad \Delta_n := \Delta(a_n).$$

The stepsize is denoted $\Delta t > 0$, and N_θ is the parameter dimension. Throughout, Δt always appears with the differential t attached, so there is no clash with the energy gap $\Delta(\cdot)$.

Assumption 2.1 (Strong log-concavity and smoothness)**.** There exist constants $0 < \alpha \le \beta < \infty$ such that

$$\alpha I \preceq \nabla^2 V(\theta) \preceq \beta I, \qquad \forall \theta \in \mathbb{R}^{N_\theta}.$$

Equivalently, $\alpha I \preceq -\nabla_\theta \nabla_\theta \log \rho_{\mathrm{post}}(\theta) \preceq \beta I$. We write the condition number

$$\kappa := \frac{\beta}{\alpha} \ge 1.$$

Assumption 2.2 (Global Hessian-Lipschitzness for stochastic burn-in)**.** For the stochastic burn-in energy estimates we additionally assume that V has a globally Lipschitz Hessian: there exists a finite constant $L_{H,\mathrm{g}} \ge 0$ such that

$$\left\|\nabla^2 V(x) - \nabla^2 V(y)\right\|_{\mathrm{op}} \le L_{H,\mathrm{g}} \|x - y\|, \qquad \forall x, y \in \mathbb{R}^{N_\theta}.$$

This assumption is used only in the stochastic burn-in energy-control results Lemmas 9.4, 10.1 and 10.4 and Theorems 10.2 and 10.5. It is not needed for the deterministic covariance bootstrap or for the post-burn-in global STL estimates.

For a Gaussian parameter $a = (m, C)$ we abbreviate

$$g(a) := \mathbb{E}_{\rho_a}[\nabla_\theta \log \rho_{\text{post}}(X)] = -\mathbb{E}_{\rho_a}[\nabla V(X)], \qquad A(a) := \mathbb{E}_{\rho_a}[\nabla^2 V(X)],$$

and the *precision residual*

$$R(a) := C^{-1} - A(a).$$

Assumption 2.1 gives $\alpha I \preceq A(a) \preceq \beta I$.

2.1 Continuous flow and deterministic schemes

The Gaussian natural gradient flow is

$$\frac{\mathrm{d}m_t}{\mathrm{d}t} = C_t g(a_t), \qquad \frac{\mathrm{d}C_t}{\mathrm{d}t} = C_t - C_t A(a_t) C_t. \tag{1}$$

The deterministic Riemannian (geodesic) scheme is

$$m_{n+1} = m_n + \Delta t\, C_n g(a_n), \qquad C_{n+1} = C_n^{1/2} \exp\bigl(\Delta t (I - C_n^{1/2} A(a_n) C_n^{1/2})\bigr) C_n^{1/2}. \tag{2}$$

The deterministic KL/Bregman scheme is

$$m_{n+1} = m_n + \Delta t\, C_n g(a_n), \qquad C_{n+1} = (1 + \Delta t)\bigl(C_n^{-1} + \Delta t\, A(a_n)\bigr)^{-1}, \tag{3}$$

or equivalently $C_{n+1}^{-1} = \frac{1}{1+\Delta t}\bigl(C_n^{-1} + \Delta t\, A(a_n)\bigr)$. We use the energy splitting $\mathcal{E} = \mathcal{E}_1 + \mathcal{E}_2$ with

$$\mathcal{E}_1(a) = \mathbb{E}_{\rho_a}[\log \rho_a] = -\tfrac{1}{2} \log \det C + \text{const.}, \qquad \mathcal{E}_2(a) = -\mathbb{E}_{\rho_a}[\log \rho_{\text{post}}].$$

2.2 Single-sample stochastic schemes

Let

$$\mathcal{F}_n := \sigma(a_0, \dots, a_n), \qquad X_n \mid \mathcal{F}_n \sim \mathcal{N}(m_n, C_n), \qquad H_n := \nabla^2 V(X_n).$$

The *raw* one-sample estimators are $b_n = \nabla \log \rho_{\text{post}}(X_n) = -\nabla V(X_n)$ and $S_n = -H_n$, and the raw stochastic schemes replace $g(a_n)$ and $A(a_n)$ by b_n and H_n in (2) and (3). We instead use the sticking-the-landing (STL) estimators. The STL score estimator is

$$\widehat{b}_n := \nabla \log \rho_{\text{post}}(X_n) - \nabla \log \rho_{a_n}(X_n) = -\nabla V(X_n) + C_n^{-1}(X_n - m_n), \tag{4}$$

and the STL residual-Hessian estimator is

$$\widehat{K}_n := C_n^{-1} + \nabla^2 \log \rho_{\text{post}}(X_n) = C_n^{-1} - H_n. \tag{5}$$

They are conditionally unbiased for the deterministic gradient and residual:

$$\mathbb{E}[\widehat{b}_n \mid \mathcal{F}_n] = g(a_n), \qquad \mathbb{E}[\widehat{K}_n \mid \mathcal{F}_n] = R(a_n). \tag{6}$$

The first identity uses $\mathbb{E}_{\rho_{a_n}}[C_n^{-1}(X_n - m_n)] = 0$; the second uses the definition of $A(a_n)$. The *STL Riemannian scheme* is

$$m_{n+1} = m_n + \Delta t\, C_n \widehat{b}_n, \qquad C_{n+1} = C_n^{1/2} \exp\bigl(\Delta t\, C_n^{1/2} \widehat{K}_n C_n^{1/2}\bigr) C_n^{1/2}, \tag{7}$$

and the *STL KL scheme* is

$$m_{n+1} = m_n + \Delta t\, C_n \widehat{b}_n, \qquad C_{n+1} = (1+\Delta t)\big((1+\Delta t)C_n^{-1} - \Delta t\, \widehat{K}_n\big)^{-1}. \tag{8}$$

Because $\widehat{K}_n = C_n^{-1} - H_n$, the covariance updates in (7) and (8) simplify to

$$C_{n+1} = C_n^{1/2} \exp\big(\Delta t(I - C_n^{1/2} H_n C_n^{1/2})\big) C_n^{1/2}, \qquad C_{n+1} = (1+\Delta t)\big(C_n^{-1} + \Delta t\, H_n\big)^{-1}, \tag{9}$$

respectively. Thus the STL covariance updates coincide with the raw Hessian-sampling updates; the improvement comes entirely from the STL score estimator and from analyzing the covariance noise in the intrinsic residual norm. In particular, since $\alpha I \preceq H_n \preceq \beta I$ almost surely, the covariance bounds proved below hold pathwise for the stochastic schemes.

2.3 Initial covariance and shorthands

We assume the initial covariance satisfies

$$\lambda_{0,\min} I \preceq C_0 \preceq \lambda_{0,\max} I,$$

and we set, as in the global theory,

$$\lambda_0 := \lambda_{0,\min}, \qquad \lambda_{\max} := \max\left\{\lambda_{0,\max}, \frac{1}{\alpha}\right\}, \qquad \log_+ x := \max\{\log x, 0\}.$$

2.4 Fisher–Rao gradient-norm convention

We use the Fisher–Rao gradient-norm convention

$$\mathsf{G}(a)^2 := g(a)^\top C g(a) + \left\| C^{1/2} R(a) C^{1/2} \right\|_{\mathrm{F}}^2. \tag{10}$$

This is the convention under which the Fisher–Rao/Wasserstein comparison and the global descent estimates are stated in the base manuscript; it absorbs harmless factors of two in the covariance component. The corresponding tangent-noise norm at $a = (m, C)$, for a tangent vector $(u, U) \in \mathbb{R}^{N_\theta} \times \mathbb{S}(N_\theta)$, is

$$\|(u, U)\|_a^2 := u^\top C^{-1} u + \left\| C^{-1/2} U C^{-1/2} \right\|_{\mathrm{F}}^2. \tag{11}$$

3 Two basic inequalities

Lemma 3.1 (Wasserstein PL inequality on Gaussians)**.** *Under Assumption 2.1, for every Gaussian parameter $a \in \mathcal{A}$,*

$$\left\|\operatorname{grad}^{W_2} \mathcal{E}(a)\right\|_{W_2}^2 \geq 2\alpha\, \Delta(a).$$

This is the Wasserstein Polyak–Łojasiewicz inequality induced by the α-strong convexity of V, restricted to the Gaussian family; see the base manuscript and [2].

Lemma 3.2 (Fisher–Rao/Wasserstein comparison)**.** *Suppose $a = (m, C)$ with $C \succeq \ell I$ for some $\ell > 0$. Then, with the convention (10),*

$$\mathsf{G}(a)^2 \geq \ell \left\|\operatorname{grad}^{W_2} \mathcal{E}(a)\right\|_{W_2}^2 \geq 2\alpha\ell\, \Delta(a).$$

Moreover,

$$\mathsf{G}(a)^2 \leq \|C\|_{\mathrm{op}} \left\|\operatorname{grad}^{W_2} \mathcal{E}(a)\right\|_{W_2}^2.$$

Proof. The Wasserstein Gaussian gradient norm is

$$\left\|\operatorname{grad}^{W_2} \mathcal{E}(a)\right\|_{W_2}^2 = \|g(a)\|^2 + \operatorname{Tr}\big(R(a)CR(a)\big),$$

and the convention (10) reads

$$\mathsf{G}(a)^2 = g(a)^\top Cg(a) + \operatorname{Tr}\big(CR(a)CR(a)\big),$$

where we used $\left\|C^{1/2}R(a)C^{1/2}\right\|_{\mathrm{F}}^2 = \operatorname{Tr}(CR(a)CR(a))$. If $C \succeq \ell I$, then $g(a)^\top Cg(a) \geq \ell \|g(a)\|^2$, and since $R(a)CR(a) \succeq 0$,

$$\operatorname{Tr}\big(CR(a)CR(a)\big) = \operatorname{Tr}\big(C\,[R(a)CR(a)]\big) \geq \ell \operatorname{Tr}\big(R(a)CR(a)\big).$$

Adding the two estimates gives $\mathsf{G}(a)^2 \geq \ell \left\|\operatorname{grad}^{W_2} \mathcal{E}(a)\right\|_{W_2}^2$, and combining with Lemma 3.1 gives the second inequality. For the upper bound, with $L_C := \|C\|_{\mathrm{op}}$,

$$g(a)^\top Cg(a) \leq L_C \|g(a)\|^2, \qquad \operatorname{Tr}\big(CR(a)CR(a)\big) = \operatorname{Tr}\big(C\,[R(a)CR(a)]\big) \leq L_C \operatorname{Tr}\big(R(a)CR(a)\big),$$

and adding gives $\mathsf{G}(a)^2 \leq L_C \left\|\operatorname{grad}^{W_2} \mathcal{E}(a)\right\|_{W_2}^2$. □

4 Continuous-time covariance bootstrap

The continuous covariance bootstrap is cleanest in precision variables.

Lemma 4.1 (Precision Duhamel formula)**.** *Let $P_t := C_t^{-1}$. Along (1),*

$$\dot{P}_t = A(a_t) - P_t, \qquad \textit{hence} \qquad P_t = e^{-t}P_0 + \int_0^t e^{-(t-s)}A(a_s)\,\mathrm{d}s.$$

Proof. Since $P_t = C_t^{-1}$, $\dot{P}_t = -P_t\dot{C}_tP_t$. Using $\dot{C}_t = C_t - C_tA(a_t)C_t$,

$$\dot{P}_t = -P_t(C_t - C_tA(a_t)C_t)P_t = -P_t + A(a_t).$$

Multiplying $\dot{P}_t + P_t = A(a_t)$ by the integrating factor e^t and integrating gives the stated formula. □

Lemma 4.2 (Continuous covariance burn-in)**.** *Under Assumption 2.1, along (1),*

$$C_t \succeq \ell(t)I, \qquad \ell(t) = \frac{\lambda_0 e^t}{1 + \beta\lambda_0(e^t - 1)}, \qquad \textit{and} \qquad C_t \preceq \lambda_{\max} I.$$

In particular $C_t \succeq \frac{1}{2\beta}I$ for all

$$t \geq T_{\mathrm{cov}}^{\mathrm{c}} := \log_+ \frac{1}{\beta\lambda_0}.$$

Proof. By Lemma 4.1 and $A(a_s) \preceq \beta I$, $P_0 \preceq \lambda_0^{-1}I$,

$$P_t \preceq \big(e^{-t}\lambda_0^{-1} + \beta(1 - e^{-t})\big)I,$$

and taking inverses yields $C_t \succeq \ell(t)I$ with $\ell(t)$ as stated. For the upper bound, $A(a_s) \succeq \alpha I$ gives $P_t \succeq e^{-t}P_0 + \alpha(1 - e^{-t})I$; since no eigenvalue of C_t crosses $1/\alpha$ from below (where $\dot{C}_t \preceq 0$) and $C_0 \preceq \lambda_{0,\max}I$, we get $C_t \preceq \lambda_{\max}I$. Finally, if $e^{-t}\lambda_0^{-1} \leq \beta$, i.e. $t \geq \log_+(1/(\beta\lambda_0))$, then $e^{-t}\lambda_0^{-1} + \beta(1 - e^{-t}) \leq 2\beta$, hence $C_t \succeq \frac{1}{2\beta}I$. □

Theorem 4.3 (Improved continuous-time global convergence). *Under Assumption 2.1, the flow* (1) *satisfies*

$$\boxed{\Delta(a_t) \le \Delta_0\big(1+\beta\lambda_0(e^t-1)\big)^{-2/\kappa}.}$$

Consequently, for any threshold $\delta \in (0,\Delta_0)$, the hitting time $T_{\mathrm{c}}^{\mathrm{glob}}(\delta) := \inf\{t \ge 0 : \Delta(a_t) \le \delta\}$ obeys

$$T_{\mathrm{c}}^{\mathrm{glob}}(\delta) = \log\left(1+\frac{(\Delta_0/\delta)^{\kappa/2}-1}{\beta\lambda_0}\right) = O\left(\log_+\frac{1}{\beta\lambda_0}+\kappa\log_+\frac{\Delta_0}{\delta}\right).$$

Proof. Along the Fisher–Rao gradient flow, $\frac{\mathrm{d}}{\mathrm{d}t}\Delta(a_t) = -\mathsf{G}(a_t)^2$. By Lemma 4.2, $C_t \succeq \ell(t)I$, so Lemma 3.2 gives $\frac{\mathrm{d}}{\mathrm{d}t}\Delta(a_t) \le -2\alpha\ell(t)\Delta(a_t)$, hence by Grönwall

$$\Delta(a_t) \le \Delta_0 \exp\left(-2\alpha\int_0^t \ell(s)\,\mathrm{d}s\right).$$

A direct computation gives $\int_0^t \ell(s)\,\mathrm{d}s = \frac{1}{\beta}\log(1+\beta\lambda_0(e^t-1))$, so

$$\Delta(a_t) \le \Delta_0\big(1+\beta\lambda_0(e^t-1)\big)^{-2\alpha/\beta} = \Delta_0\big(1+\beta\lambda_0(e^t-1)\big)^{-2/\kappa}.$$

Solving $\Delta_0(1+\beta\lambda_0(e^t-1))^{-2/\kappa} = \delta$ for t gives the displayed hitting time, and the big-O form follows from $\log(1+xy) \le \log_+ x + \log_+ y + \log 2$ for $x,y \ge 0$ together with $\log((\Delta_0/\delta)^{\kappa/2}) = \frac{\kappa}{2}\log(\Delta_0/\delta)$. □

5 Riemannian scheme: spectral bounds and two-sided covariance bootstrap

Lemma 5.1 (Riemannian covariance bootstrap). *Assume Assumption 2.1, $\lambda_{0,\min}I \preceq C_0 \preceq \lambda_{0,\max}I$, and*

$$0 < \Delta t \le \min\left\{1, \frac{1}{\beta\lambda_{\max}}\right\}.$$

Then $C_n \preceq \lambda_{\max}I$ for all $n \ge 0$. Define

$$L_0 := \min\left\{\lambda_0, \frac{1}{2\beta}\right\}, \qquad L_{n+1} := \frac{e^{\Delta t}L_n}{1+(e-1)\Delta t\,\beta L_n}.$$

Then $C_n \succeq L_n I$ for all $n \ge 0$, and with

$$y_0 := \frac{1}{\beta L_0}, \qquad y_\infty := \frac{(e-1)\Delta t}{e^{\Delta t}-1} \in [1, e-1) \subset [1,2),$$

one has $y_n := (\beta L_n)^{-1} = y_\infty + (y_0 - y_\infty)e^{-n\Delta t}$. Hence $C_n \succeq \frac{1}{2\beta}I$ for all

$$n \ge N_{\mathrm{cov},-}^{\mathrm{R}} := \left\lceil \frac{1}{\Delta t}\log_+\frac{y_0-y_\infty}{2-y_\infty}\right\rceil = O\left(\frac{1}{\Delta t}\log_+\frac{1}{\beta\lambda_0}\right).$$

Moreover, with $z_0 := \frac{1}{\alpha\lambda_{0,\max}}$ and $z_\infty := \frac{\Delta t}{e^{\Delta t}-1} \in (\frac{1}{e-1}, 1]$, one has $z_n := \lambda_{\min}(C_n^{-1})/\alpha \ge z_\infty + (z_0 - z_\infty)e^{-n\Delta t}$, hence $C_n \preceq \frac{2}{\alpha}I$ for all

$$n \ge N_{\mathrm{cov},+}^{\mathrm{R}} := \left\lceil \frac{1}{\Delta t}\log_+\frac{z_\infty - z_0}{z_\infty - 1/2}\right\rceil = O\left(\frac{1}{\Delta t}\log_+(\alpha\lambda_{0,\max})\right).$$

Proof. Write $B_n := C_n^{1/2} A(a_n) C_n^{1/2}$, so that the covariance update reads $C_{n+1} = C_n^{1/2} \exp(\Delta t(I - B_n)) C_n^{1/2}$ and $C_{n+1}^{-1} = C_n^{-1/2} \exp(\Delta t(B_n - I)) C_n^{-1/2}$.

Upper bound. Using $e^X \succeq I + X$ for symmetric X,

$$C_{n+1}^{-1} \succeq C_n^{-1/2}\big[I + \Delta t(B_n - I)\big]C_n^{-1/2} = (1-\Delta t)C_n^{-1} + \Delta t\, A(a_n),$$

using $C_n^{-1/2} B_n C_n^{-1/2} = A(a_n)$. Inductively assume $C_n \preceq \lambda_{\max} I$; then $C_n^{-1} \succeq \lambda_{\max}^{-1} I$ and $A(a_n) \succeq \alpha I \succeq \lambda_{\max}^{-1} I$ (since $\lambda_{\max} \geq 1/\alpha$). As $\Delta t \leq 1$,

$$C_{n+1}^{-1} \succeq \big[(1-\Delta t)\lambda_{\max}^{-1} + \Delta t\,\alpha\big] I \succeq \lambda_{\max}^{-1} I,$$

hence $C_{n+1} \preceq \lambda_{\max} I$; the base case is the hypothesis on C_0.

Lower envelope. Since $A(a_n) \preceq \beta I$ and $C_n \preceq \lambda_{\max} I$, $0 \preceq B_n \preceq \beta\lambda_{\max} I$, so $0 \preceq \Delta t\, B_n \preceq I$ (because $\Delta t \leq 1/(\beta\lambda_{\max})$). The scalar inequality $e^x \leq 1 + (e-1)x$ on $[0,1]$ gives, by functional calculus, $\exp(\Delta t\, B_n) \preceq I + (e-1)\Delta t\, B_n$, hence

$$C_{n+1}^{-1} = e^{-\Delta t} C_n^{-1/2} \exp(\Delta t\, B_n) C_n^{-1/2} \preceq e^{-\Delta t}\big[C_n^{-1} + (e-1)\Delta t\, A(a_n)\big].$$

If $C_n \succeq L_n I$ then $C_n^{-1} \preceq L_n^{-1} I$ and $A(a_n) \preceq \beta I$, so

$$C_{n+1}^{-1} \preceq e^{-\Delta t}\big(L_n^{-1} + (e-1)\Delta t\,\beta\big) I,$$

i.e. $C_{n+1} \succeq L_{n+1} I$. With $y_n = (\beta L_n)^{-1}$ the recurrence becomes the affine recursion $y_{n+1} = e^{-\Delta t}(y_n + (e-1)\Delta t)$, whose solution is $y_n = y_\infty + (y_0 - y_\infty)e^{-n\Delta t}$ with $y_\infty = (e-1)\Delta t/(e^{\Delta t} - 1)$. For $\Delta t \in (0,1]$ the map $\Delta t \mapsto y_\infty$ is continuous with limit $e-1$ as $\Delta t \to 0$ and value 1 at $\Delta t = 1$, and is monotone, so $y_\infty \in [1, e-1) \subset [1,2)$. The condition $C_n \succeq \frac{1}{2\beta} I$ is $y_n \leq 2$, i.e. $(y_0 - y_\infty)e^{-n\Delta t} \leq 2 - y_\infty$, giving $N_{\mathrm{cov},-}^{\mathrm{R}}$. Since $L_0 = \min\{\lambda_0, 1/(2\beta)\}$, this is $O(\Delta t^{-1} \log_+(1/(\beta\lambda_0)))$.

Upper envelope. Using $e^X \succeq I + X$ again, $C_{n+1}^{-1} \succeq e^{-\Delta t}(C_n^{-1} + \Delta t\, A(a_n)) \succeq e^{-\Delta t}(C_n^{-1} + \Delta t\,\alpha I)$. Hence $p_n := \lambda_{\min}(C_n^{-1})$ obeys $p_{n+1} \geq e^{-\Delta t}(p_n + \Delta t\,\alpha)$, and with $z_n = p_n/\alpha$ the equality recursion has solution $z_n = z_\infty + (z_0 - z_\infty)e^{-n\Delta t}$ with $z_\infty = \Delta t/(e^{\Delta t} - 1)$. For $\Delta t \in (0,1]$, $z_\infty \in (\frac{1}{e-1}, 1]$, in particular $z_\infty > 1/2$. The condition $C_n \preceq \frac{2}{\alpha} I$ is $z_n \geq 1/2$, giving $N_{\mathrm{cov},+}^{\mathrm{R}}$, which is $O(\Delta t^{-1} \log_+(\alpha\lambda_{0,\max}))$. □

Theorem 5.2 (Improved global hitting time, Riemannian scheme)**.** *Under Assumption 2.1, with* $0 < \Delta t \leq \min\{1, 1/(\beta\lambda_{\max})\}$, *the hitting time* $N_{\mathrm{R}}^{\mathrm{glob}}(\delta) := \inf\{n \geq 0 : \Delta(a_n) \leq \delta\}$ *of* (2) *satisfies, for any* $\delta \in (0, \Delta_0)$,

$$\boxed{N_{\mathrm{R}}^{\mathrm{glob}}(\delta) = O\left(\frac{1}{\Delta t}\log_+ \frac{1}{\beta\lambda_0} + \frac{\kappa}{\Delta t}\log_+ \frac{\Delta_0}{\delta}\right).}$$

Proof. The Riemannian smoothness estimate of the base manuscript (with geodesic smoothness constant $L = \beta\lambda_{\max}$) gives

$$\Delta(a_{n+1}) \leq \Delta(a_n) - \Delta t\Big(1 - \tfrac{\beta\lambda_{\max}\Delta t}{2}\Big)\mathsf{G}(a_n)^2.$$

If $C_n \succeq L_n I$, Lemma 3.2 gives $\mathsf{G}(a_n)^2 \geq 2\alpha L_n \Delta(a_n)$, hence

$$\Delta(a_{n+1}) \leq \big(1 - \alpha L_n \Delta t(2 - \beta\lambda_{\max}\Delta t)\big)\Delta(a_n) \leq (1 - \alpha L_n \Delta t)\Delta(a_n),$$

using $2 - \beta\lambda_{\max}\Delta t \geq 1$. After $N_{\mathrm{cov},-}^{\mathrm{R}}$ iterations, Lemma 5.1 gives $L_n \geq 1/(2\beta)$, so the per-step factor becomes $1 - \frac{\Delta t}{2\kappa}$, and reducing the gap from Δ_0 to δ needs $O(\frac{\kappa}{\Delta t}\log_+ \frac{\Delta_0}{\delta})$ further iterations. □

6 KL/Bregman scheme: spectral bounds and exact two-sided covariance bootstrap

Lemma 6.1 (KL covariance bootstrap)**.** *Assume Assumption 2.1 and* $\lambda_{0,\min} I \preceq C_0 \preceq \lambda_{0,\max} I$. *Along* (3) *(and, pathwise, along the stochastic KL update with* $A(a_n)$ *replaced by* H_n*),* $C_n \preceq \lambda_{\max} I$ *for all* $n \geq 0$*. Define*

$$L_0 := \min\left\{\lambda_0, \frac{1}{2\beta}\right\}, \qquad L_{n+1} = \frac{(1+\Delta t)L_n}{1+\Delta t\,\beta L_n},$$

so that $C_n \succeq L_n I$*, with the explicit solution*

$$L_n = \frac{1}{\beta\big[1 + (\frac{1}{\beta L_0} - 1)(1+\Delta t)^{-n}\big]}.$$

Hence $C_n \succeq \frac{1}{2\beta} I$ *for all*

$$n \geq N^{\mathrm{KL}}_{\mathrm{cov},-} := \left\lceil \frac{\log_+\big(\frac{1}{\beta L_0} - 1\big)}{\log(1+\Delta t)} \right\rceil = O\left(\frac{1}{\Delta t}\log_+ \frac{1}{\beta\lambda_0}\right),$$

and $C_n \preceq \frac{2}{\alpha} I$ *for all*

$$n \geq N^{\mathrm{KL}}_{\mathrm{cov},+} := \left\lceil \frac{\log 2}{\log(1+\Delta t)} \right\rceil = O\left(\frac{1}{\Delta t}\right).$$

Proof. The update is $C_{n+1}^{-1} = \frac{1}{1+\Delta t}(C_n^{-1} + \Delta t\, A(a_n))$ with $\alpha I \preceq A(a_n) \preceq \beta I$ (the same bound holds pathwise for H_n).

Upper bound. If $C_n \preceq \lambda_{\max} I$ then $C_n^{-1} \succeq \lambda_{\max}^{-1} I$ and $A(a_n) \succeq \alpha I \succeq \lambda_{\max}^{-1} I$, so $C_{n+1}^{-1} \succeq \frac{1}{1+\Delta t}(\lambda_{\max}^{-1} + \Delta t \lambda_{\max}^{-1}) I = \lambda_{\max}^{-1} I$, i.e. $C_{n+1} \preceq \lambda_{\max} I$.

Lower envelope. If $C_n \succeq L_n I$ then $C_n^{-1} \preceq L_n^{-1} I$, $A(a_n) \preceq \beta I$, so $C_{n+1}^{-1} \preceq \frac{1}{1+\Delta t}(L_n^{-1} + \Delta t \beta) I$, i.e. $C_{n+1} \succeq L_{n+1} I$. With $y_n = (\beta L_n)^{-1}$ the recursion is $y_{n+1} = \frac{y_n + \Delta t}{1+\Delta t} = 1 + \frac{y_n - 1}{1+\Delta t}$, so $y_n = 1 + (y_0 - 1)(1+\Delta t)^{-n}$, which is the stated closed form. The condition $C_n \succeq \frac{1}{2\beta} I$ is $y_n \leq 2$, i.e. $(y_0 - 1)(1+\Delta t)^{-n} \leq 1$, giving $N^{\mathrm{KL}}_{\mathrm{cov},-}$ (here $y_0 = 1/(\beta L_0)$, and $\log_+(y_0 - 1)$ covers the case $y_0 \leq 2$).

Upper envelope. With $A(a_n) \succeq \alpha I$, $p_n := \lambda_{\min}(C_n^{-1})$ obeys $p_{n+1} \geq \frac{p_n + \Delta t\,\alpha}{1+\Delta t}$, and with $z_n = p_n/\alpha$ the equality recursion is $z_{n+1} = 1 + \frac{z_n - 1}{1+\Delta t}$, so $z_n = 1 + (z_0 - 1)(1+\Delta t)^{-n}$. If $z_0 \geq 1$ then $z_n \geq 1 \geq \frac{1}{2}$ for all n. If $0 \leq z_0 < 1$ then $z_n \geq \frac{1}{2}$ once $(1 - z_0)(1+\Delta t)^{-n} \leq \frac{1}{2}$, for which $(1+\Delta t)^{-n} \leq \frac{1}{2}$ (i.e. $n \geq N^{\mathrm{KL}}_{\mathrm{cov},+}$) suffices since $1 - z_0 \leq 1$. In either case $C_n \preceq \frac{2}{\alpha} I$ for $n \geq N^{\mathrm{KL}}_{\mathrm{cov},+}$. □

Theorem 6.2 (Improved global hitting time, KL scheme)**.** *Under Assumption 2.1, with* $0 < \Delta t \leq 1/(\beta\lambda_{\max})$*, the hitting time* $N^{\mathrm{glob}}_{\mathrm{KL}}(\delta) := \inf\{n \geq 0 : \Delta(a_n) \leq \delta\}$ *of* (3) *satisfies, for any* $\delta \in (0, \Delta_0)$*,*

$$\boxed{N^{\mathrm{glob}}_{\mathrm{KL}}(\delta) = O\left(\frac{1}{\Delta t}\log_+ \frac{1}{\beta\lambda_0} + \frac{\kappa}{\Delta t}\log_+ \frac{\Delta_0}{\delta}\right).}$$

Proof. The KL one-step estimate of the base manuscript gives $\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n) - \frac{1}{\Delta t}\mathrm{KL}(\rho_{a_n} \| \rho_{a_{n+1}})$ and, with the dynamic lower bound $C_n \succeq L_n I$ in place of the frozen one,

$$\mathrm{KL}(\rho_{a_n} \| \rho_{a_{n+1}}) \geq \frac{L_n \Delta t^2}{4(1+\Delta t)(1+\Delta t \beta \lambda_{\max})} \left\|\mathrm{grad}^{W_2} \mathcal{E}(a_n)\right\|^2_{W_2}.$$

By Lemma 3.1, $\left\|\operatorname{grad}^{W_2}\mathcal{E}(a_n)\right\|_{W_2}^2 \ge 2\alpha\Delta(a_n)$, hence

$$\Delta(a_{n+1}) \le \left(1 - \frac{\alpha L_n \Delta t}{2(1+\Delta t)(1+\Delta t\beta\lambda_{\max})}\right)\Delta(a_n).$$

After $N^{\mathrm{KL}}_{\mathrm{cov},-}$ iterations, Lemma 6.1 gives $L_n \ge 1/(2\beta)$; and since $\Delta t \le 1/(\beta\lambda_{\max})$ we have $\Delta t \le 1$ and $\Delta t\beta\lambda_{\max} \le 1$, so $(1+\Delta t)(1+\Delta t\beta\lambda_{\max}) \le 4$. Thus the per-step factor is at most $1 - \frac{\Delta t}{16\kappa}$, and reducing the gap to δ needs $O(\frac{\kappa}{\Delta t}\log_+\frac{\Delta_0}{\delta})$ further iterations. □

7 Local input and deterministic three-stage complexity

The local convergence theory is proved after equilibrium whitening in the base manuscript. We record it here in original coordinates, through intrinsic equilibrium norms, so that it can be chained to the global estimates.

Define the equilibrium-normalized variables associated with $a = (m, C)$ by $\widehat{m} := C_\star^{-1/2}(m - m_\star)$, $\widehat{C} := C_\star^{-1/2} C C_\star^{-1/2}$, so that $a_\star$ corresponds to $(0, I)$. The original-coordinate local norms are

$$\|a - a_\star\|_\star^2 := \|\widehat{m}\|^2 + \tfrac{1}{2}\left\|\widehat{C} - I\right\|_{\mathrm{F}}^2, \qquad \|a - a_\star\|_{\star,\Delta t}^2 := \|\widehat{m}\|^2 + \tfrac{1+\Delta t}{2}\left\|\widehat{C} - I\right\|_{\mathrm{F}}^2. \tag{12}$$

Let $\widehat{V}(z) := V(m_\star + C_\star^{1/2} z)$, let $0 < \alpha_\star \le 1 \le \beta_\star$ satisfy $\alpha_\star I \preceq \nabla^2 \widehat{V}(z) \preceq \beta_\star I$ for all z, and set $\kappa_\star := \beta_\star/\alpha_\star$ and

$$\Gamma := \min\left\{\beta_\star - 1,\ N_\theta\Big(3 + \tfrac{4}{\sqrt{\pi}}(1 + \log\kappa_\star)\Big)\right\}. \tag{13}$$

Assumption 7.1 (Deterministic local convergence, original coordinates)**.** With (12)–(13), the base manuscript provides local radii

$$r_{\mathrm{c}} := \min\Big\{\tfrac{1}{2\sqrt{2}}, \tfrac{1}{135\beta_\star N_\theta^{5/2}(4+\Gamma)}\Big\}, \quad r_{\mathrm{R}} := \min\Big\{\tfrac{1}{2\sqrt{2}}, \tfrac{1}{(4+\Gamma)\,148\beta_\star^2 N_\theta^{5/2}}\Big\},$$

$$r_{\mathrm{KL}} := \min\Big\{\tfrac{1}{2\sqrt{2}}, \tfrac{1}{(4+2\Delta t+\Gamma)\,216\beta_\star^2 N_\theta^{5/2}}\Big\},$$

such that the following hold. If $\|a_0 - a_\star\|_\star \le r_{\mathrm{c}}$, then along (1),

$$\|a_t - a_\star\|_\star^2 \le e^{-t/(4+\Gamma)}\,\|a_0 - a_\star\|_\star^2.$$

If $0 < \Delta t \le \frac{2}{4+\Gamma}$ and $\|a_0 - a_\star\|_\star \le r_{\mathrm{R}}$, then along (2),

$$\|a_n - a_\star\|_\star \le \left(1 - \tfrac{\Delta t}{2(4+\Gamma)}\right)^n \|a_0 - a_\star\|_\star.$$

If $0 < \Delta t \le \frac{2}{4+\Gamma}$ and $\|a_0 - a_\star\|_{\star,\Delta t} \le r_{\mathrm{KL}}$, then along (3),

$$\|a_n - a_\star\|_{\star,\Delta t} \le \left(1 - \tfrac{\Delta t}{2(4+2\Delta t+\Gamma)}\right)^n \|a_0 - a_\star\|_{\star,\Delta t}.$$

Energy-to-local-norm conversion. Let $\chi_\star := \left\|C_\star^{-1}\right\|_{\mathrm{op}}$; since $C_\star^{-1} = \mathbb{E}_{\rho_{a_\star}}[\nabla^2 V]$, Assumption 2.1 gives $\chi_\star \le \beta$. If $C \preceq \lambda_{\max} I$, then $\widehat{C} \preceq \widehat{\lambda}_{\max} I$ with $\widehat{\lambda}_{\max} := \chi_\star \lambda_{\max}$, and $\left\|\widehat{C} - I\right\|_{\mathrm{F}}^2 \le (1+\sqrt{\widehat{\lambda}_{\max}})^2 \left\|\widehat{C}^{1/2} - I\right\|_{\mathrm{F}}^2$. Define

$$A_0 := \max\Big\{1, \tfrac{(1+\sqrt{\widehat{\lambda}_{\max}})^2}{2}\Big\}, \qquad A_{\Delta t} := \max\Big\{1, \tfrac{(1+\Delta t)(1+\sqrt{\widehat{\lambda}_{\max}})^2}{2}\Big\}.$$

Writing $\widehat{\rho}_a$ for the pushforward of ρ_a under $\theta \mapsto C_\star^{-1/2}(\theta - m_\star)$, one has $W_2^2(\widehat{\rho}_a, \widehat{\rho}_{a_\star}) = \|\widehat{m}\|^2 + \left\|\widehat{C}^{1/2} - I\right\|_{\mathrm{F}}^2$, and $W_2^2(\widehat{\rho}_a, \widehat{\rho}_{a_\star}) \le \chi_\star W_2^2(\rho_a, \rho_{a_\star})$ since $\left\|C_\star^{-1/2}\right\|_{\mathrm{op}} = \sqrt{\chi_\star}$. Together with the Wasserstein strong convexity $\Delta(a) \ge \frac{\alpha}{2} W_2^2(\rho_a, \rho_{a_\star})$, this yields

$$\|a - a_\star\|_\star^2 \le \frac{2\chi_\star A_0}{\alpha}\Delta(a), \qquad \|a - a_\star\|_{\star,\Delta t}^2 \le \frac{2\chi_\star A_{\Delta t}}{\alpha}\Delta(a).$$

Define the local energy thresholds

$$\Delta_{\mathrm{c}} := \frac{\alpha r_{\mathrm{c}}^2}{2\chi_\star A_0}, \qquad \Delta_{\mathrm{R}} := \frac{\alpha r_{\mathrm{R}}^2}{2\chi_\star A_0}, \qquad \Delta_{\mathrm{KL}} := \frac{\alpha r_{\mathrm{KL}}^2}{2\chi_\star A_{\Delta t}}, \tag{14}$$

so that $\Delta(a) \le \Delta_{\mathrm{c}} \Rightarrow \|a - a_\star\|_\star \le r_{\mathrm{c}}$, and similarly for the Riemannian and KL thresholds.

Theorem 7.2 (Deterministic three-stage complexity)**.** *Assume Assumptions 2.1 and 7.1 and $\lambda_{0,\min} I \preceq C_0 \preceq \lambda_{0,\max} I$. Let Δ_{c}, Δ_{R}, Δ_{KL} be as in (14). For sufficiently small $\epsilon > 0$:*

1. *Continuous-time flow (1):*

$$T_{\mathrm{c}}(\epsilon) = O\bigg(\log_+ \frac{1}{\beta\lambda_0} + \kappa \log_+ \frac{\Delta_0}{\Delta_{\mathrm{c}}} + (4+\Gamma)\log\frac{1}{\epsilon}\bigg)$$

suffices for $\|a_T - a_\star\|_\star^2 \le \epsilon$.

2. *Riemannian scheme (2), with $0 < \Delta t \le \min\{1/(\beta\lambda_{\max}),\, 2/(4+\Gamma)\}$:*

$$N_{\mathrm{R}}(\epsilon) = O\bigg(\frac{1}{\Delta t}\log_+ \frac{1}{\beta\lambda_0} + \frac{\kappa}{\Delta t}\log_+ \frac{\Delta_0}{\Delta_{\mathrm{R}}} + \frac{4+\Gamma}{\Delta t}\log\frac{1}{\epsilon}\bigg)$$

suffices for $\|a_N - a_\star\|_\star^2 \le \epsilon$.

3. *KL scheme (3), with $0 < \Delta t \le \min\{1/(\beta\lambda_{\max}),\, 2/(4+\Gamma)\}$:*

$$N_{\mathrm{KL}}(\epsilon) = O\bigg(\frac{1}{\Delta t}\log_+ \frac{1}{\beta\lambda_0} + \frac{\kappa}{\Delta t}\log_+ \frac{\Delta_0}{\Delta_{\mathrm{KL}}} + \frac{4+2\Delta t+\Gamma}{\Delta t}\log\frac{1}{\epsilon}\bigg)$$

suffices for $\|a_N - a_\star\|_{\star,\Delta t}^2 \le \epsilon$.

Proof. Continuous time. By Theorem 4.3 with $\delta = \Delta_{\mathrm{c}}$, after time $O(\log_+ \frac{1}{\beta\lambda_0} + \kappa \log_+ \frac{\Delta_0}{\Delta_{\mathrm{c}}})$ one has $\Delta(a_t) \le \Delta_{\mathrm{c}}$, hence $\|a_t - a_\star\|_\star \le r_{\mathrm{c}}$ by (14). The local estimate of Assumption 7.1 then needs $O((4+\Gamma)\log\frac{1}{\epsilon})$ further time to reach $\|a_t - a_\star\|_\star^2 \le \epsilon$ (since r_{c} is independent of ϵ).

Riemannian. By Theorem 5.2 with $\delta = \Delta_{\mathrm{R}}$, after $O(\frac{1}{\Delta t}\log_+ \frac{1}{\beta\lambda_0} + \frac{\kappa}{\Delta t}\log_+ \frac{\Delta_0}{\Delta_{\mathrm{R}}})$ iterations one has $\Delta(a_n) \le \Delta_{\mathrm{R}}$, hence $\|a_n - a_\star\|_\star \le r_{\mathrm{R}}$. The local estimate then needs $O(\frac{4+\Gamma}{\Delta t}\log\frac{1}{\epsilon})$ further iterations. The KL case is identical using Theorem 6.2 and the KL local estimate. The big-O forms use $-\log(1-x) \ge x$. $\square$

8 Transport bootstrap: Wasserstein burn-in followed by Fisher–Rao tail

The deterministic covariance bootstrap in the preceding sections shows that pure Fisher–Rao/natural-gradient dynamics has only logarithmic dependence on the initial lower covariance. This is already a substantial improvement over the frozen-bound analysis. However, when C_0 is severely underdispersed, even the logarithmic term $\log_+(1/(\beta\lambda_0))$ is a genuine Fisher–Rao warm-up cost. This section records a clean way to remove that term: use Wasserstein transport only to lift the covariance to the curvature scale, and then switch permanently to Fisher–Rao.

The main point is not to run a Wasserstein–Fisher–Rao splitting at every iteration. Instead, we separate the roles of the two geometries:

Wasserstein transport lifts an underdispersed covariance; Fisher–Rao performs the global-to-local and local c

This avoids an adaptive transport schedule and avoids the cost of re-evaluating a Fisher–Rao step after a Wasserstein half-step at every iteration.

8.1 Covariance-scale comparison

Let

$$A(a) := \mathbb{E}_{\rho_a}[\nabla^2 V(X)] = -\mathbb{E}_{\rho_a}[\nabla_\theta^2 \log \rho_{\text{post}}(X)], \qquad \alpha I \preceq A(a) \preceq \beta I.$$

For the Fisher–Rao covariance equation,

$$\dot{C} = C - CAC,$$

if $\mu_i(t)$ is an eigenvalue of C_t with normalized eigenvector $u_i(t)$, then

$$\dot{\mu}_i(t) = \mu_i(t) - \mu_i(t)^2 h_i(t) = \mu_i(t)(1 - \mu_i(t)h_i(t)), \qquad h_i(t) := u_i(t)^\top A(a_t)u_i(t) \in [\alpha, \beta].$$

Thus, in the underdispersed regime $\mu_i \ll 1/\beta$,

$$\dot{\mu}_i(t) \approx \mu_i(t),$$

so the Fisher–Rao covariance growth is multiplicative. Starting from $\mu_i(0) = \lambda_0 \ll 1/\beta$, reaching the scale $1/\beta$ necessarily takes time of order $\log(1/(\beta\lambda_0))$.

By contrast, the unit-strength Gaussian Bures–Wasserstein flow is

$$\dot{m}_t = g(a_t), \qquad \dot{C}_t = 2I - C_t A(a_t) - A(a_t)C_t. \tag{15}$$

For the same eigenpair,

$$\dot{\mu}_i(t) = 2(1 - \mu_i(t)h_i(t)).$$

Thus, when $\mu_i \ll 1/\beta$,

$$\dot{\mu}_i(t) \approx 2,$$

so Wasserstein covariance growth is additive and scale-independent. This is the precise reason transport is better suited to the covariance warm-up.

The same comparison also explains why transport should not be kept forever. In a flat direction with curvature $h_i = \alpha$, growing from $1/\beta$ to the target scale $1/\alpha$ takes Wasserstein time of order

$$\frac{1}{\alpha} - \frac{1}{\beta} = O\left(\frac{1}{\alpha}\right),$$

whereas Fisher–Rao, whose growth is approximately multiplicative until the target scale, takes only

$$O\left(\log \frac{\beta}{\alpha}\right) = O(\log \kappa).$$

Thus the clean covariance-scale picture is

$$\boxed{\text{Wasserstein is best for } 0 \to O(1/\beta), \qquad \text{Fisher–Rao is best for shape calibration } 1/\beta \to 1/\alpha.}$$

This comparison is a covariance-scale dominance statement, not a pointwise claim that one energy-dissipation rate dominates the other at every state.

8.2 Continuous-time Wasserstein bootstrap

Lemma 8.1 (Wasserstein covariance bootstrap)**.** *Assume Assumption 2.1 and $\lambda_{0,\min} I \preceq C_0 \preceq \lambda_{0,\max} I$. Let $\lambda_0 := \lambda_{0,\min}$ and $\lambda_{\max} := \max\{\lambda_{0,\max}, 1/\alpha\}$. Along the Gaussian Wasserstein flow (15),*

$$C_t \succeq \ell_W(t) I, \qquad \ell_W(t) := \frac{1}{\beta} + \left(\lambda_0 - \frac{1}{\beta}\right) e^{-2\beta t},$$

while $C_t \preceq \lambda_{\max} I$. Consequently, for any fixed $c \in (0,1)$, if

$$T_W^{\mathrm{c}}(c) := \begin{cases} 0, & \lambda_0 \geq c/\beta, \\ \dfrac{1}{2\beta} \log \dfrac{1-\beta\lambda_0}{1-c}, & \lambda_0 < c/\beta, \end{cases}$$

then

$$C_{T_W^{\mathrm{c}}(c)} \succeq \frac{c}{\beta} I.$$

In particular, for fixed c, the Wasserstein covariance warm-up time is bounded by a constant multiple of $1/\beta$ and has no logarithmic dependence on λ_0.

Proof. Let $\mu_i(t)$ be an eigenvalue of C_t with normalized eigenvector $u_i(t)$. Along (15),

$$\dot{\mu}_i(t) = u_i(t)^\top (2I - C_t A(a_t) - A(a_t) C_t) u_i(t) = 2\big(1 - \mu_i(t) h_i(t)\big),$$

where $h_i(t) := u_i(t)^\top A(a_t) u_i(t) \in [\alpha, \beta]$. Hence

$$\dot{\mu}_i(t) \geq 2(1 - \beta \mu_i(t)).$$

Let ℓ_W solve

$$\dot{\ell}_W(t) = 2(1 - \beta \ell_W(t)), \qquad \ell_W(0) = \lambda_0.$$

The solution is

$$\ell_W(t) = \frac{1}{\beta} + \left(\lambda_0 - \frac{1}{\beta}\right) e^{-2\beta t}.$$

The scalar comparison argument gives $\mu_i(t) \geq \ell_W(t)$ for every i, and hence $C_t \succeq \ell_W(t) I$.

For the upper bound, if $\mu_i(t) \geq 1/\alpha$, then

$$\dot{\mu}_i(t) = 2(1 - \mu_i(t) h_i(t)) \leq 2(1 - \alpha \mu_i(t)) \leq 0.$$

Therefore no eigenvalue can cross the barrier $1/\alpha$ from below, and any eigenvalue that starts above $1/\alpha$ is pushed downward. Since $C_0 \preceq \lambda_{0,\max} I$, this gives $C_t \preceq \lambda_{\max} I$.

Finally, if $\lambda_0 < c/\beta$, the condition $\ell_W(t) \ge c/\beta$ is equivalent to

$$e^{-2\beta t} \le \frac{1-c}{1-\beta\lambda_0},$$

which gives the stated expression for $T_W^{\mathrm{c}}(c)$. If $\lambda_0 \ge c/\beta$, no Wasserstein warm-up time is needed. $\square$

Theorem 8.2 (Continuous transport-bootstrap natural gradient)**.** *Assume Assumptions 2.1 and 7.1. Fix $c \in (0,1)$. Run the Gaussian Wasserstein flow (15) until time $T_W^{\mathrm{c}}(c)$ from Lemma 8.1, and then switch permanently to the Fisher–Rao flow (1). Let Δ_{c} be the local energy threshold in (14). Then, for sufficiently small $\epsilon > 0$,*

$$T^{\mathrm{c}}_{W\to\mathrm{FR}}(\epsilon; c) \le T_W^{\mathrm{c}}(c) + \frac{\kappa}{2c}\log_+\frac{\Delta_0}{\Delta_{\mathrm{c}}} + (4+\Gamma)\log\frac{r_{\mathrm{c}}^2}{\epsilon}.$$

In particular, for any fixed $c \in (0,1)$,

$$\boxed{T^{\mathrm{c}}_{W\to\mathrm{FR}}(\epsilon; c) = O\left(\frac{1}{\beta} + \kappa\log_+\frac{\Delta_0}{\Delta_{\mathrm{c}}} + (4+\Gamma)\log\frac{1}{\epsilon}\right).}$$

Thus the pure-Fisher–Rao covariance-burn-in term $\log_+(1/(\beta\lambda_0))$ is replaced by a Wasserstein warm-start time of order $1/\beta$.

Proof. During the Wasserstein stage, $\mathcal{E}$ is decreasing because (15) is the Bures–Wasserstein gradient flow of $\mathcal{E}$ on the Gaussian manifold. Hence, at the switching state a_{b},

$$\Delta(a_{\mathrm{b}}) \le \Delta_0, \qquad C_{\mathrm{b}} \succeq \frac{c}{\beta} I.$$

After the switch, the Fisher–Rao/Wasserstein comparison in Lemma 3.2 gives

$$\mathsf{G}(a_t)^2 \ge \frac{c}{\beta}\left\|\operatorname{grad}^{W_2}\mathcal{E}(a_t)\right\|_{W_2}^2 \ge \frac{2\alpha c}{\beta}\Delta(a_t) = \frac{2c}{\kappa}\Delta(a_t),$$

as long as the Fisher–Rao trajectory is in the global phase. Since $\dot{\Delta}(t) = -\mathsf{G}(a_t)^2$ along Fisher–Rao,

$$\dot{\Delta}(t) \le -\frac{2c}{\kappa}\Delta(t).$$

Therefore, after an additional time

$$\frac{\kappa}{2c}\log_+\frac{\Delta_0}{\Delta_{\mathrm{c}}},$$

we have $\Delta(a_t) \le \Delta_{\mathrm{c}}$. By (14), this implies $\|a_t - a_\star\|_\star \le r_{\mathrm{c}}$. The local continuous-time estimate in Assumption 7.1 then gives

$$\|a_{t+s} - a_\star\|_\star^2 \le e^{-s/(4+\Gamma)} r_{\mathrm{c}}^2.$$

Taking $s = (4+\Gamma)\log(r_{\mathrm{c}}^2/\epsilon)$ proves the explicit bound. The big-O form follows from Lemma 8.1, which gives $T_W^{\mathrm{c}}(c) \le (2\beta)^{-1}\log(1/(1-c))$ when c is fixed. $\square$

8.3 One-step discrete transport bootstrap

The discrete version is even cleaner. A single Bures–Wasserstein forward–backward step creates a covariance floor of size equal to the transport stepsize.

Given $a = (m, C)$ and a transport stepsize $\eta > 0$, define

$$g := g(a), \qquad H := \mathbb{E}_{\rho_a}[\nabla_\theta^2 \log \rho_{\mathrm{post}}(X)] = -A(a), \qquad M := I + \eta H = I - \eta A(a), \tag{16}$$

$$m^+ := m + \eta g, \qquad \widetilde{C} := MCM,$$
$$C^+ := \frac{1}{2}\left(\widetilde{C} + 2\eta I + \left[\widetilde{C}(\widetilde{C} + 4\eta I)\right]^{1/2}\right). \tag{17}$$

The matrix square root is well-defined because $\widetilde{C}$ commutes with $\widetilde{C} + 4\eta I$.

Lemma 8.3 (One-step Wasserstein covariance floor)**.** *Assume Assumption 2.1, $C \preceq \lambda_{\max} I$ with $\lambda_{\max} \geq 1/\alpha$, and $0 < \eta \leq 1/\beta$. Then the Wasserstein step (16)–(17) satisfies*

$$C^+ \succeq \eta I, \qquad C^+ \preceq \lambda_{\max} I.$$

Moreover, the standard Bures–Wasserstein forward–backward descent inequality gives

$$\mathcal{E}(m^+, C^+) \leq \mathcal{E}(m, C).$$

Proof. Let

$$\psi_\eta(x) := \frac{1}{2}\left(x + 2\eta + \sqrt{x(x + 4\eta)}\right), \qquad x \geq 0.$$

The covariance update (17) applies ψ_η to the eigenvalues of $\widetilde{C}$. Since ψ_η is increasing and $\psi_\eta(0) = \eta$, every eigenvalue of C^+ is at least η. Thus $C^+ \succeq \eta I$.

For the upper bound, since $0 < \eta \leq 1/\beta$ and $\alpha I \preceq A(a) \preceq \beta I$,

$$0 \preceq M = I - \eta A(a) \preceq (1 - \eta\alpha) I.$$

Therefore

$$\widetilde{C} = MCM \preceq (1 - \eta\alpha)^2 \lambda_{\max} I.$$

Because $\lambda_{\max} \geq 1/\alpha$, we have $\eta\alpha \geq \eta/\lambda_{\max}$, hence

$$(1 - \eta\alpha)^2 \lambda_{\max} \leq \left(1 - \frac{\eta}{\lambda_{\max}}\right)^2 \lambda_{\max}.$$

The scalar identity

$$\psi_\eta\left(\left(1 - \frac{\eta}{L}\right)^2 L\right) = L, \qquad 0 \leq \eta \leq L,$$

with $L = \lambda_{\max}$ gives $C^+ \preceq \lambda_{\max} I$.

Finally, (16)–(17) is exactly the Bures–Wasserstein forward–backward step for the splitting $\mathcal{E} = \mathcal{E}_1 + \mathcal{E}_2$, with the entropy part treated backward. Under $\eta \leq 1/\beta$, the standard forward–backward descent inequality yields $\mathcal{E}(m^+, C^+) \leq \mathcal{E}(m, C)$. □

Theorem 8.4 (Discrete transport-bootstrap natural gradient)**.** *Assume Assumptions 2.1 and 7.1 and $\lambda_{0,\min} I \preceq C_0 \preceq \lambda_{0,\max} I$. Fix $c \in (0, 1]$ and set*

$$\eta := \frac{c}{\beta}.$$

If $\lambda_0 < c/\beta$, first perform one Wasserstein forward–backward step (16)–(17); otherwise skip this step. Then run either the Riemannian scheme (2) or the KL scheme (3) with Fisher–Rao stepsize Δt.

For the Riemannian scheme, if

$$0 < \Delta t \le \min\left\{\frac{1}{\beta\lambda_{\max}}, \frac{2}{4+\Gamma}\right\},$$

then, for sufficiently small $\epsilon > 0$,

$$\boxed{N_{W\to \mathrm{R}}(\epsilon; c) = \mathbf{1}_{\{\lambda_0 < c/\beta\}} + O\left(\frac{\kappa}{c\Delta t}\log_+ \frac{\Delta_0}{\Delta_{\mathrm{R}}} + \frac{4+\Gamma}{\Delta t}\log\frac{1}{\epsilon}\right).}$$

For the KL scheme, under the same stepsize condition,

$$\boxed{N_{W\to \mathrm{KL}}(\epsilon; c) = \mathbf{1}_{\{\lambda_0 < c/\beta\}} + O\left(\frac{\kappa}{c\Delta t}\log_+ \frac{\Delta_0}{\Delta_{\mathrm{KL}}} + \frac{4+2\Delta t+\Gamma}{\Delta t}\log\frac{1}{\epsilon}\right).}$$

Thus, for fixed c, the discrete covariance-burn-in term $\Delta t^{-1}\log_+(1/(\beta\lambda_0))$ is replaced by one Wasserstein bootstrap step.

Proof. If the Wasserstein step is skipped, then $C_0 \succeq (c/\beta)I$ already. If the step is taken, Lemma 8.3 gives

$$C_{\mathrm{b}} \succeq \frac{c}{\beta} I, \qquad C_{\mathrm{b}} \preceq \lambda_{\max} I, \qquad \Delta(a_{\mathrm{b}}) \le \Delta_0.$$

Here a_{b} denotes the post-bootstrap state. Hence the subsequent Fisher–Rao stage starts with the fixed covariance floor c/β and the same upper bound $\lambda_{\max}$.

For the Riemannian scheme, the one-step descent estimate used in Theorem 5.2, now with the lower covariance floor c/β, gives

$$\Delta_{n+1} \le \left(1 - \frac{\alpha c}{\beta}\Delta t(2 - \beta\lambda_{\max}\Delta t)\right)\Delta_n.$$

Since $\Delta t \le 1/(\beta\lambda_{\max})$, we have $2 - \beta\lambda_{\max}\Delta t \ge 1$, so

$$\Delta_{n+1} \le \left(1 - \frac{c\Delta t}{\kappa}\right)\Delta_n.$$

Thus reaching Δ_{R} from a gap at most Δ_0 requires $O((\kappa/(c\Delta t))\log_+(\Delta_0/\Delta_{\mathrm{R}}))$ Riemannian iterations. Once $\Delta \le \Delta_{\mathrm{R}}$, (14) places the iterate in the local ball, and Assumption 7.1 gives an additional $O(((4+\Gamma)/\Delta t)\log(1/\epsilon))$ iterations.

For the KL scheme, the one-step estimate used in Theorem 6.2 gives

$$\Delta_{n+1} \le \left(1 - \frac{\alpha(c/\beta)\Delta t}{2(1+\Delta t)(1+\Delta t\beta\lambda_{\max})}\right)\Delta_n.$$

Under $\Delta t \le 1$ and $\Delta t\beta\lambda_{\max} \le 1$, the denominator is bounded by 8, so the global localization phase takes $O((\kappa/(c\Delta t))\log_+(\Delta_0/\Delta_{\mathrm{KL}}))$ iterations. The KL local theorem in Assumption 7.1 then gives the stated local term. ∎

Remark 8.5 (What the transport bootstrap does and does not solve). The transport bootstrap removes the small-initial-covariance burn-in from the deterministic global theory. It does not, by itself, improve the explicit local discretization factor $(4+\Gamma)/\Delta t$ for the Riemannian scheme or $(4+2\Delta t+\Gamma)/\Delta t$ for the KL scheme. Thus the discrete local stiffness issue is separate from the global covariance warm-up issue. To obtain a fully ideal discrete local term of order $\Gamma \log(1/\epsilon)$, one would need an additional locally implicit, proximal, or exponential-stabilized Fisher–Rao discretization.

Remark 8.6 (Practical switching rule). The theorem suggests a particularly simple deterministic implementation. Pick a constant $c \in (0,1]$, for example $c = 1/2$, perform one Wasserstein step with $\eta = c/\beta$ if $\beta\lambda_{\min}(C_0) < c$, and then switch permanently to the preferred Fisher–Rao discretization. Equivalently, one may monitor

$$r_n := \beta\lambda_{\min}(C_n)$$

and switch once $r_n \geq c$. With the forward–backward Wasserstein bootstrap, this typically happens after one step by Lemma 8.3.

9 Intrinsic sticking-the-landing variance

We now turn to the stochastic schemes (7) and (8). The deterministic gradient and its single-sample STL estimate are

$$\operatorname{grad}\mathcal{E}(a) = \big(-Cg(a),\, -CR(a)C\big), \qquad \widehat{\operatorname{grad}}\mathcal{E}(a;X) = \big(-C\widehat{b},\, -C\widehat{K}C\big),$$

where $\widehat{b} = -\nabla V(X) + C^{-1}(X-m)$ and $\widehat{K} = C^{-1} - \nabla^2 V(X)$. The induced STL tangent noise is

$$\xi(a;X) := \widehat{\operatorname{grad}}\mathcal{E}(a;X) - \operatorname{grad}\mathcal{E}(a) = \big(-C\varepsilon_b(a;X),\, -C\varepsilon_R(a;X)C\big), \tag{18}$$

with the centered noises

$$\varepsilon_b(a;X) := \widehat{b} - g(a), \qquad \varepsilon_R(a;X) := \widehat{K} - R(a) = -\big(\nabla^2 V(X) - A(a)\big).$$

Recall the intrinsic Hessian fluctuation

$$\Psi(a) := \mathbb{E}_{X\sim\rho_a}\left[\left\|C^{1/2}\big(\nabla^2 V(X) - A(a)\big)C^{1/2}\right\|_{\mathrm{F}}^2\right] = \mathbb{E}_{X\sim\rho_a}\left[\left\|C^{1/2}\varepsilon_R(a;X)C^{1/2}\right\|_{\mathrm{F}}^2\right]. \tag{19}$$

Lemma 9.1 (Intrinsic STL variance bound)**.** *Under Assumption 2.1, for every $a = (m,C) \in \mathcal{A}$ the STL noises are conditionally centered, $\mathbb{E}[\varepsilon_b(a;X)] = 0$ and $\mathbb{E}[\varepsilon_R(a;X)] = 0$, and*

$$\mathbb{E}\big[\varepsilon_b(a;X)^\top C\,\varepsilon_b(a;X)\big] \leq \left\|C^{1/2}R(a)C^{1/2}\right\|_{\mathrm{F}}^2 + \Psi(a). \tag{20}$$

Consequently, with the tangent norm (11) and the gradient-norm convention (10),

$$\boxed{\mathbb{E}\big[\|\xi(a;X)\|_a^2\big] \leq \mathsf{G}(a)^2 + 2\Psi(a).} \tag{21}$$

Proof. Centering. $\mathbb{E}[\widehat{b}] = -\mathbb{E}[\nabla V(X)] + C^{-1}\mathbb{E}[X-m] = g(a)$ and $\mathbb{E}[\widehat{K}] = C^{-1} - A(a) = R(a)$, so both noises are centered.

Covariance component. Directly from (19), $\mathbb{E}[\|C^{1/2}\varepsilon_R(a;X)C^{1/2}\|_{\mathrm{F}}^2] = \Psi(a)$.

Mean component. Let $F(X) := -\nabla V(X) + C^{-1}(X-m)$, so $\varepsilon_b = F(X) - \mathbb{E}F(X)$ and the Jacobian $\nabla F(X) = C^{-1} - \nabla^2 V(X)$ is symmetric. We use the scalar Gaussian Poincaré inequality:

for $X \sim \mathcal{N}(m, C)$ and $\phi \in C^1$, $\operatorname{Var}(\phi(X)) \leq \mathbb{E}[\nabla\phi(X)^\top C \nabla\phi(X)]$. Writing $\phi_i(X) := (C^{1/2}F(X))_i$, so that $C^{1/2}\varepsilon_b$ has components $\phi_i(X) - \mathbb{E}\phi_i(X)$, we have $\nabla\phi_i(X) = (C^{1/2}\nabla F(X))_{i\cdot}^\top$ (the i-th row of $C^{1/2}\nabla F(X)$, transposed). Summing the scalar Poincaré bound over i,

$$\begin{aligned}\mathbb{E}[\varepsilon_b^\top C \varepsilon_b] = \sum_i \operatorname{Var}(\phi_i(X)) \leq \sum_i \mathbb{E}[\nabla\phi_i^\top C \nabla\phi_i] &= \mathbb{E}\big[\operatorname{Tr}\big(C^{1/2}\nabla F(X)\, C\, \nabla F(X) C^{1/2}\big)\big] \\ &= \mathbb{E}\big[\big\|C^{1/2}\nabla F(X) C^{1/2}\big\|_{\mathrm{F}}^2\big],\end{aligned}$$

where the last equality uses the symmetry of $\nabla F(X)$. Now $C^{1/2}\nabla F(X)C^{1/2} = C^{1/2}(C^{-1} - \nabla^2 V(X))C^{1/2} = C^{1/2}R(a)C^{1/2} - C^{1/2}(\nabla^2 V(X) - A(a))C^{1/2}$, since $C^{-1} - \nabla^2 V(X) = R(a) - (\nabla^2 V(X) - A(a))$. Expanding the squared Frobenius norm and taking expectation, the cross term vanishes because $\mathbb{E}[\nabla^2 V(X) - A(a)] = 0$ and $C^{1/2}R(a)C^{1/2}$ is deterministic. Hence

$$\mathbb{E}[\varepsilon_b^\top C \varepsilon_b] \leq \big\|C^{1/2}R(a)C^{1/2}\big\|_{\mathrm{F}}^2 + \mathbb{E}\big[\big\|C^{1/2}(\nabla^2 V(X) - A(a))C^{1/2}\big\|_{\mathrm{F}}^2\big] = \big\|C^{1/2}R(a)C^{1/2}\big\|_{\mathrm{F}}^2 + \Psi(a),$$

which is (20).

Assembly. With the tangent noise (18) and the norm (11), a direct computation (as in Remark 9.2 of the displayed identity below) gives

$$\|\xi(a; X)\|_a^2 = \varepsilon_b^\top C \varepsilon_b + \big\|C^{1/2}\varepsilon_R C^{1/2}\big\|_{\mathrm{F}}^2.$$

Taking expectations and inserting (20) and the covariance component,

$$\mathbb{E}\big[\|\xi(a; X)\|_a^2\big] \leq \big\|C^{1/2}R(a)C^{1/2}\big\|_{\mathrm{F}}^2 + \Psi(a) + \Psi(a) \leq \mathsf{G}(a)^2 + 2\Psi(a),$$

using $\|C^{1/2}R(a)C^{1/2}\|_{\mathrm{F}}^2 \leq \mathsf{G}(a)^2$ from (10). $\square$

Remark 9.2 (The tangent-noise identity). With $u = -C\varepsilon_b$ and $U = -C\varepsilon_R C$ in (11), $u^\top C^{-1} u = \varepsilon_b^\top C \varepsilon_b$ and $\|C^{-1/2}UC^{-1/2}\|_{\mathrm{F}}^2 = \|C^{1/2}\varepsilon_R C^{1/2}\|_{\mathrm{F}}^2$, which is the identity used above.

Lemma 9.3 (Global bound for Ψ after two-sided burn-in)**.** *Under Assumption 2.1, if $\frac{1}{2\beta}I \preceq C \preceq \frac{2}{\alpha}I$, then*

$$\boxed{\Psi(a) \leq 4N_\theta(\kappa - 1)^2.}$$

Proof. Since $\alpha I \preceq \nabla^2 V(X) \preceq \beta I$ and $\alpha I \preceq A(a) \preceq \beta I$, $\big\|\nabla^2 V(X) - A(a)\big\|_{\mathrm{op}} \leq \beta - \alpha$, hence $\big\|\nabla^2 V(X) - A(a)\big\|_{\mathrm{F}} \leq \sqrt{N_\theta}(\beta - \alpha)$. Therefore

$$\big\|C^{1/2}(\nabla^2 V(X) - A(a))C^{1/2}\big\|_{\mathrm{F}} \leq \|C\|_{\mathrm{op}} \big\|\nabla^2 V(X) - A(a)\big\|_{\mathrm{F}} \leq \frac{2}{\alpha}\sqrt{N_\theta}(\beta - \alpha).$$

Squaring gives $\Psi(a) \leq \frac{4}{\alpha^2}N_\theta(\beta - \alpha)^2 = 4N_\theta(\kappa - 1)^2$. $\square$

Lemma 9.4 (Hessian-Lipschitz control of intrinsic Hessian fluctuation)**.** *Assume Assumption 2.2. Let $X \sim \mathcal{N}(m, C)$ and let $B \succeq 0$ be any deterministic symmetric positive semidefinite matrix. Then*

$$\mathbb{E}\left[\big\|B^{1/2}\big(\nabla^2 V(X) - \mathbb{E}\nabla^2 V(X)\big)B^{1/2}\big\|_{\mathrm{F}}^2\right] \leq L_{H,\mathrm{g}}^2 \, \|B\|_{\mathrm{op}} \operatorname{Tr}(B) \operatorname{Tr}(C).$$

In particular, taking $B = C$ gives the covariance-dependent intrinsic bound

$$\boxed{\Psi(a) \le L_{H,\mathrm{g}}^2 \, \|C\|_{\mathrm{op}} \, \big(\operatorname{Tr} C\big)^2.} \tag{22}$$

Consequently, if $C \preceq \Lambda I$, then

$$\boxed{\Psi(a) \le N_\theta^2 L_{H,\mathrm{g}}^2 \Lambda^3.} \tag{23}$$

Proof. Write $X = m + C^{1/2}Z$ with $Z \sim \mathcal{N}(0, I)$ and define the matrix-valued function

$$F(Z) := B^{1/2}\nabla^2 V(m + C^{1/2}Z)B^{1/2}.$$

The Gaussian Poincaré inequality applied componentwise to F gives

$$\mathbb{E}\,\|F(Z) - \mathbb{E}F(Z)\|_{\mathrm{F}}^2 \le \mathbb{E}\,\|D_Z F(Z)\|_{\mathrm{HS}}^2\,,$$

where the Hilbert–Schmidt norm on the right is taken for the linear map $D_Z F(Z) : \mathbb{R}^{N_\theta} \to \mathbb{S}(N_\theta)$ with the Frobenius norm on matrices. For any unit vector u,

$$D_Z F(Z)[u] = B^{1/2}\, D(\nabla^2 V)(X)[C^{1/2}u]\, B^{1/2}.$$

By Assumption 2.2,

$$\left\| D(\nabla^2 V)(X)[C^{1/2}u] \right\|_{\mathrm{op}} \le L_{H,\mathrm{g}} \left\| C^{1/2}u \right\|.$$

Therefore

$$\|D_Z F(Z)[u]\|_{\mathrm{F}} \le \left\|B^{1/2}\right\|_{\mathrm{op}} \, L_{H,\mathrm{g}} \left\|C^{1/2}u\right\| \, \left\|B^{1/2}\right\|_{\mathrm{F}} = L_{H,\mathrm{g}} \, \|B\|_{\mathrm{op}}^{1/2} \, \operatorname{Tr}(B)^{1/2} \, \left\|C^{1/2}u\right\|.$$

Summing the square of this bound over an orthonormal basis $\{e_i\}_{i=1}^{N_\theta}$ gives

$$\|D_Z F(Z)\|_{\mathrm{HS}}^2 \le L_{H,\mathrm{g}}^2 \, \|B\|_{\mathrm{op}} \operatorname{Tr}(B) \sum_{i=1}^{N_\theta} \left\|C^{1/2}e_i\right\|^2 = L_{H,\mathrm{g}}^2 \, \|B\|_{\mathrm{op}} \operatorname{Tr}(B)\operatorname{Tr}(C).$$

This proves the first claim. The special case $B = C$ gives (22). If $C \preceq \Lambda I$, then $\|C\|_{\mathrm{op}} \le \Lambda$ and $\operatorname{Tr} C \le N_\theta \Lambda$, which gives (23). ☐

Remark 9.5 (Pathwise covariance bootstrap). The covariance updates of the STL schemes (9) are the deterministic updates with $A(a_n)$ replaced by the sampled Hessian $H_n = \nabla^2 V(X_n)$, which satisfies $\alpha I \preceq H_n \preceq \beta I$ almost surely. Therefore the lower and upper covariance envelopes of Lemma 5.1 (Riemannian) and Lemma 6.1 (KL) hold *pathwise* for the stochastic schemes. In particular, defining the two-sided burn-in times

$$N_{\mathrm{cov}}^{\mathrm{R}} := \max\{N_{\mathrm{cov},-}^{\mathrm{R}}, N_{\mathrm{cov},+}^{\mathrm{R}}\} = O\Big(\tfrac{1}{\Delta t}\log_+ \tfrac{1}{\beta\lambda_0} + \tfrac{1}{\Delta t}\log_+(\alpha\lambda_{0,\max})\Big)\,,$$

$$N_{\mathrm{cov}}^{\mathrm{KL}} := \max\{N_{\mathrm{cov},-}^{\mathrm{KL}}, N_{\mathrm{cov},+}^{\mathrm{KL}}\} = O\Big(\tfrac{1}{\Delta t}\log_+ \tfrac{1}{\beta\lambda_0}\Big)\,,$$

we have, for the respective scheme,

$$\tfrac{1}{2\beta} I \preceq C_n \preceq \tfrac{2}{\alpha} I \qquad \text{for all } n \ge N_{\mathrm{cov}}^{\bullet},\ \bullet \in \{\mathrm{R}, \mathrm{KL}\},$$

almost surely. By Lemma 9.3, the post-burn-in fluctuation levels

$$\Psi_\star^{\mathrm{R}} := \sup_{n \ge N_{\mathrm{cov}}^{\mathrm{R}}} \Psi(a_n), \qquad \Psi_\star^{\mathrm{KL}} := \sup_{n \ge N_{\mathrm{cov}}^{\mathrm{KL}}} \Psi(a_n)$$

satisfy $\Psi_\star^{\mathrm{R}}, \Psi_\star^{\mathrm{KL}} \le 4N_\theta(\kappa - 1)^2$.

10 Stochastic energy control during covariance burn-in

The pathwise covariance bootstrap in Remark 9.5 controls the covariance of the stochastic STL schemes during the first stage, but it does not by itself control the energy gap. This section supplies the missing estimate. The key new input is the global Hessian-Lipschitz bound in Assumption 2.2, which converts the intrinsic Hessian fluctuation into a covariance-dependent quantity through Lemma 9.4. As a result, the stochastic error accumulated during covariance burn-in is of order

$$\Delta t^2 N_{\mathrm{cov}}\, \beta N_\theta^2 L_{H,\mathrm{g}}^2 \lambda_{\max}^4,$$

with every constant explicit.

10.1 Riemannian STL burn-in energy estimate

Let L_n^{R} denote the deterministic lower covariance envelope from Lemma 5.1, and let $N_{\mathrm{cov}}^{\mathrm{R}}$ be the two-sided pathwise covariance burn-in time from Remark 9.5. Define

$$B_{\mathrm{burn}}^{\mathrm{R}} := \beta N_\theta^2 L_{H,\mathrm{g}}^2 \lambda_{\max}^4.$$

Lemma 10.1 (Riemannian STL burn-in one-step estimate). *Assume Assumptions 2.1 and 2.2 and*

$$0 < \Delta t \le \frac{1}{2\beta\lambda_{\max}}.$$

Along the STL Riemannian scheme (7), for every $0 \le n < N_{\mathrm{cov}}^{\mathrm{R}}$,

$$\boxed{\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \le \big(1 - \alpha L_n^{\mathrm{R}} \Delta t\big)\Delta_n + B_{\mathrm{burn}}^{\mathrm{R}} \Delta t^2.}$$

The same bound is valid for all $n \ge 0$ as long as $C_n \succeq L_n^{\mathrm{R}} I$ and $C_n \preceq \lambda_{\max} I$.

Proof. The stochastic Riemannian step is the geodesic step

$$a_{n+1} = \mathrm{Exp}_{a_n}\big(-\Delta t\, \widehat{\mathrm{grad}}\mathcal{E}(a_n)\big),$$

as in Proposition 12.2. Since H_n satisfies $\alpha I \preceq H_n \preceq \beta I$ almost surely, the same upper covariance envelope as in Lemma 5.1 holds pathwise; the geodesic segment generated by any partial step $s\Delta t$, $s \in [0,1]$, also satisfies the upper bound $C \preceq \lambda_{\max} I$. Hence the geodesic smoothness constant of $\mathcal{E}$ along the step is

$$L_{\mathrm{sm}} = \beta\lambda_{\max}.$$

Writing $\xi_n := \widehat{\mathrm{grad}}\mathcal{E}(a_n) - \mathrm{grad}\,\mathcal{E}(a_n)$ and using geodesic smoothness,

$$\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \le \Delta_n - \Delta t \mathsf{G}(a_n)^2 + \frac{\beta\lambda_{\max}\Delta t^2}{2}\mathbb{E}\big[\|\mathrm{grad}\,\mathcal{E}(a_n) + \xi_n\|_{a_n}^2 \mid \mathcal{F}_n\big].$$

The STL estimator is conditionally unbiased, so the cross term vanishes:

$$\mathbb{E}\big[\|\mathrm{grad}\,\mathcal{E}(a_n) + \xi_n\|_{a_n}^2 \mid \mathcal{F}_n\big] = \mathsf{G}(a_n)^2 + \mathbb{E}[\|\xi_n\|_{a_n}^2 \mid \mathcal{F}_n].$$

By Lemma 9.1,

$$\mathbb{E}[\|\xi_n\|_{a_n}^2 \mid \mathcal{F}_n] \le \mathsf{G}(a_n)^2 + 2\Psi(a_n).$$

Therefore

$$\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \le \Delta_n - \Delta t(1-\beta\lambda_{\max}\Delta t)\mathsf{G}(a_n)^2 + \beta\lambda_{\max}\Delta t^2\Psi(a_n).$$

Since $\Delta t \le 1/(2\beta\lambda_{\max})$, $1-\beta\lambda_{\max}\Delta t \ge 1/2$. By Lemma 3.2 and $C_n \succeq L_n^{\mathrm{R}} I$,

$$\mathsf{G}(a_n)^2 \ge 2\alpha L_n^{\mathrm{R}}\Delta_n.$$

Thus

$$\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \le \Delta_n - \alpha L_n^{\mathrm{R}}\Delta t\,\Delta_n + \beta\lambda_{\max}\Delta t^2\Psi(a_n).$$

Finally, since $C_n \preceq \lambda_{\max} I$, Lemma 9.4 gives $\Psi(a_n) \le N_\theta^2 L_{H,\mathrm{g}}^2\lambda_{\max}^3$, and hence $\beta\lambda_{\max}\Delta t^2\Psi(a_n) \le \beta N_\theta^2 L_{H,\mathrm{g}}^2\lambda_{\max}^4\Delta t^2 = B_{\mathrm{burn}}^{\mathrm{R}}\Delta t^2$. This proves the claim. $\square$

Theorem 10.2 (Riemannian STL energy at covariance burn-in)**.** *Assume Assumptions 2.1 and 2.2 and $0 < \Delta t \le \frac{1}{2\beta\lambda_{\max}}$. Define, for $0 \le j \le N$, $\Pi_{j,N}^{\mathrm{R}} := \prod_{i=j}^{N-1}(1-\alpha L_i^{\mathrm{R}}\Delta t)$, $\Pi_{N,N}^{\mathrm{R}} := 1$. Then*

$$\boxed{\mathbb{E}\Delta_{N_{\mathrm{cov}}^{\mathrm{R}}} \le \Pi_{0,N_{\mathrm{cov}}^{\mathrm{R}}}^{\mathrm{R}}\Delta_0 + B_{\mathrm{burn}}^{\mathrm{R}}\Delta t^2 \sum_{j=0}^{N_{\mathrm{cov}}^{\mathrm{R}}-1}\Pi_{j+1,N_{\mathrm{cov}}^{\mathrm{R}}}^{\mathrm{R}}.} \tag{24}$$

In particular,

$$\boxed{\mathbb{E}\Delta_{N_{\mathrm{cov}}^{\mathrm{R}}} \le \Delta_0 + \beta N_\theta^2 L_{H,\mathrm{g}}^2\lambda_{\max}^4\Delta t^2 N_{\mathrm{cov}}^{\mathrm{R}}.} \tag{25}$$

Proof. By Lemma 10.1, for $0 \le n < N_{\mathrm{cov}}^{\mathrm{R}}$, $\mathbb{E}\Delta_{n+1} \le (1-\alpha L_n^{\mathrm{R}}\Delta t)\mathbb{E}\Delta_n + B_{\mathrm{burn}}^{\mathrm{R}}\Delta t^2$. Iterating this scalar recursion gives (24). Since every factor in $\Pi_{j,N}^{\mathrm{R}}$ lies in $[0,1]$, the product term is at most Δ_0 and the sum is at most $N_{\mathrm{cov}}^{\mathrm{R}}$, which gives (25). $\square$

10.2 KL STL burn-in energy estimate

Let L_n^{KL} denote the lower covariance envelope from Lemma 6.1, and let $N_{\mathrm{cov}}^{\mathrm{KL}}$ be the two-sided pathwise covariance burn-in time from Remark 9.5. Define $B_{\mathrm{burn}}^{\mathrm{KL}} := 3\beta N_\theta^2 L_{H,\mathrm{g}}^2\lambda_{\max}^4$. For the KL proof we need one deterministic lower bound that controls the Fisher–Rao gradient norm directly.

Lemma 10.3 (KL deterministic step controls the Fisher–Rao norm)**.** *Let $\bar{a}_{n+1}$ be the deterministic KL update (3) from a_n, and assume $C_n \preceq \lambda_{\max} I$. Then*

$$D(a_n, \bar{a}_{n+1}) = \mathrm{KL}(\rho_{a_n} \,\|\, \rho_{\bar{a}_{n+1}}) \ge \frac{\Delta t^2}{4(1+\Delta t)(1+\Delta t\beta\lambda_{\max})}\mathsf{G}(a_n)^2.$$

Proof. Let $B_n := C_n^{1/2}A(a_n)C_n^{1/2}$ and let $b_{i,n}$ be its eigenvalues. Since $A(a_n) \preceq \beta I$ and $C_n \preceq \lambda_{\max} I$, we have $0 \le b_{i,n} \le \beta\lambda_{\max}$. The deterministic covariance update gives $\bar{C}_{n+1}^{-1}C_n \sim \frac{I+\Delta t B_n}{1+\Delta t}$, so its eigenvalues are $r_{i,n} = (1+\Delta t b_{i,n})/(1+\Delta t)$. The covariance part of $D(a_n, \bar{a}_{n+1})$ is $D^C = \frac{1}{2}\sum_i(r_{i,n} - \log r_{i,n} - 1)$. Using $x - \log x - 1 \ge (x-1)^2/(2\max\{x,1\})$ and $\max_i r_{i,n} \le (1+\Delta t\beta\lambda_{\max})/(1+\Delta t)$,

$$D^C \ge \frac{\Delta t^2}{4(1+\Delta t)(1+\Delta t\beta\lambda_{\max})}\sum_i (b_{i,n}-1)^2 = \frac{\Delta t^2}{4(1+\Delta t)(1+\Delta t\beta\lambda_{\max})}\left\|C_n^{1/2}R(a_n)C_n^{1/2}\right\|_{\mathrm{F}}^2.$$

The mean part is

$$\begin{aligned} D^m &= \frac{1}{2}(\Delta t\, C_n g(a_n))^\top \bar{C}_{n+1}^{-1}(\Delta t\, C_n g(a_n)) = \frac{\Delta t^2}{2(1+\Delta t)}g(a_n)^\top\big(C_n + \Delta t\, C_n A(a_n) C_n\big)g(a_n) \\ &\ge \frac{\Delta t^2}{2(1+\Delta t)}\,g(a_n)^\top C_n g(a_n) \ge \frac{\Delta t^2}{4(1+\Delta t)(1+\Delta t\beta\lambda_{\max})}\,g(a_n)^\top C_n g(a_n). \end{aligned}$$

Adding the mean and covariance lower bounds gives the claim by the definition (10) of $\mathsf{G}(a_n)^2$. $\square$

Lemma 10.4 (KL STL burn-in one-step estimate)**.** *Assume Assumptions 2.1 and 2.2 and $0 < \Delta t \leq \frac{1}{32\beta\lambda_{\max}}$. Along the STL KL scheme (8), for every $0 \leq n < N_{\mathrm{cov}}^{\mathrm{KL}}$,*

$$\boxed{\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \leq \left(1 - \tfrac{\alpha L_n^{\mathrm{KL}} \Delta t}{16}\right)\Delta_n + B_{\mathrm{burn}}^{\mathrm{KL}} \Delta t^2.}$$

The same bound is valid for all $n \geq 0$ as long as $C_n \succeq L_n^{\mathrm{KL}} I$ and $C_n \preceq \lambda_{\max} I$.

Proof. Let $\bar{a}_{n+1}$ be the deterministic KL update from a_n. We first bound the stochastic excess $\mathbb{E}[\mathcal{E}(a_{n+1}) - \mathcal{E}(\bar{a}_{n+1}) \mid \mathcal{F}_n]$.

The entropy term is nonpositive in expectation. Indeed, with $P_{n+1} = C_{n+1}^{-1}$ and $\bar{P}_{n+1} = \bar{C}_{n+1}^{-1}$, the STL covariance update gives $\mathbb{E}[P_{n+1} \mid \mathcal{F}_n] = \bar{P}_{n+1}$, and concavity of $\log\det$ yields $\mathbb{E}[\mathcal{E}_1(a_{n+1}) - \mathcal{E}_1(\bar{a}_{n+1}) \mid \mathcal{F}_n] \leq 0$.

For the potential term, relative smoothness of $\mathcal{E}_2$ on the covariance range $C \preceq \lambda_{\max} I$ gives

$$\mathcal{E}_2(a_{n+1}) \leq \mathcal{E}_2(\bar{a}_{n+1}) + \langle \nabla \mathcal{E}_2(\bar{a}_{n+1}), a_{n+1} - \bar{a}_{n+1} \rangle + \beta\lambda_{\max} D(a_{n+1}, \bar{a}_{n+1}).$$

The mean part of the linear term is centered because $m_{n+1} - \bar{m}_{n+1} = \Delta t\, C_n \varepsilon_b$ and $\mathbb{E}[\varepsilon_b \mid \mathcal{F}_n] = 0$. For the covariance part, set $W_n := C_{n+1}^{-1} - \bar{C}_{n+1}^{-1} = -\frac{\Delta t}{1+\Delta t}\varepsilon_R$ and $Y_n := \bar{C}_{n+1}^{1/2} W_n \bar{C}_{n+1}^{1/2}$, so $\bar{C}_{n+1}^{-1/2} C_{n+1} \bar{C}_{n+1}^{-1/2} = (I + Y_n)^{-1}$. Since $\left\|\bar{C}_{n+1}\right\|_{\mathrm{op}} \leq \lambda_{\max}$ and $\|\varepsilon_R\|_{\mathrm{op}} \leq \beta - \alpha \leq \beta$, the stepsize condition gives $\|Y_n\|_{\mathrm{op}} \leq \frac{\Delta t}{1+\Delta t}\lambda_{\max}\beta \leq \frac{1}{32} \leq \frac{1}{2}$. Moreover, by Lemma 9.4 with $B = \bar{C}_{n+1}$ and $C = C_n$,

$$\mathbb{E}[\|Y_n\|_{\mathrm{F}}^2 \mid \mathcal{F}_n] \leq \Delta t^2 L_{H,\mathrm{g}}^2 \left\|\bar{C}_{n+1}\right\|_{\mathrm{op}} \operatorname{Tr}(\bar{C}_{n+1}) \operatorname{Tr}(C_n) \leq N_\theta^2 L_{H,\mathrm{g}}^2 \lambda_{\max}^3 \Delta t^2.$$

Writing $C_{n+1} - \bar{C}_{n+1} = \bar{C}_{n+1}^{1/2}(-Y_n + Y_n^2(I + Y_n)^{-1})\bar{C}_{n+1}^{1/2}$, the term linear in Y_n vanishes after conditioning. Since $\left\|\bar{C}_{n+1}^{1/2} A(\bar{a}_{n+1}) \bar{C}_{n+1}^{1/2}\right\|_{\mathrm{op}} \leq \beta\lambda_{\max}$ and $\left\|(I + Y_n)^{-1}\right\|_{\mathrm{op}} \leq 2$, the covariance part of the linear potential term is bounded by $\beta\lambda_{\max}\mathbb{E}[\|Y_n\|_{\mathrm{F}}^2 \mid \mathcal{F}_n] \leq \beta N_\theta^2 L_{H,\mathrm{g}}^2 \lambda_{\max}^4 \Delta t^2$.

It remains to bound $D(a_{n+1}, \bar{a}_{n+1})$. Its mean part is $D^m = \frac{\Delta t^2}{2}\varepsilon_b^\top C_n \bar{C}_{n+1}^{-1} C_n \varepsilon_b \leq \Delta t^2 \varepsilon_b^\top C_n \varepsilon_b$, because $C_n \bar{C}_{n+1}^{-1} C_n = \frac{1}{1+\Delta t}(C_n + \Delta t\, C_n A(a_n) C_n) \preceq \frac{1+\Delta t\beta\lambda_{\max}}{1+\Delta t} C_n \preceq C_n$ under $\Delta t\beta\lambda_{\max} \leq 1$. By Lemma 9.1 and Lemma 9.4, $\mathbb{E}[D^m \mid \mathcal{F}_n] \leq \Delta t^2(\mathsf{G}(a_n)^2 + \Psi(a_n)) \leq \Delta t^2 \mathsf{G}(a_n)^2 + N_\theta^2 L_{H,\mathrm{g}}^2 \lambda_{\max}^3 \Delta t^2$. The covariance part satisfies $D^C \leq \|Y_n\|_{\mathrm{F}}^2$ for $\|Y_n\|_{\mathrm{op}} \leq 1/2$, hence $\mathbb{E}[D^C \mid \mathcal{F}_n] \leq N_\theta^2 L_{H,\mathrm{g}}^2 \lambda_{\max}^3 \Delta t^2$. Therefore $\mathbb{E}[D(a_{n+1}, \bar{a}_{n+1}) \mid \mathcal{F}_n] \leq \Delta t^2 \mathsf{G}(a_n)^2 + 2N_\theta^2 L_{H,\mathrm{g}}^2 \lambda_{\max}^3 \Delta t^2$.

Combining the entropy, linear potential, and relative-smoothness terms,

$$\mathbb{E}[\mathcal{E}(a_{n+1}) - \mathcal{E}(\bar{a}_{n+1}) \mid \mathcal{F}_n] \leq \beta\lambda_{\max}\Delta t^2 \mathsf{G}(a_n)^2 + 3\beta N_\theta^2 L_{H,\mathrm{g}}^2 \lambda_{\max}^4 \Delta t^2. \tag{26}$$

Now use deterministic KL descent. By the three-point argument for the deterministic KL step, $\Delta(\bar{a}_{n+1}) \leq \Delta_n - \frac{1}{\Delta t} D(a_n, \bar{a}_{n+1})$. By Lemma 10.3, $\frac{1}{\Delta t} D(a_n, \bar{a}_{n+1}) \geq c_{\mathrm{KL}} \Delta t\, \mathsf{G}(a_n)^2$ with $c_{\mathrm{KL}} := \frac{1}{4(1+\Delta t)(1+\Delta t\beta\lambda_{\max})}$. The stepsize condition gives $\Delta t \leq 1$ and $\Delta t\beta\lambda_{\max} \leq 1/32$, hence $c_{\mathrm{KL}} \geq 1/9$, and $\beta\lambda_{\max}\Delta t^2 \leq \frac{1}{32}\Delta t \leq \frac{1}{2} c_{\mathrm{KL}} \Delta t$. Adding (26) to the deterministic descent leaves at least $\frac{1}{2} c_{\mathrm{KL}} \Delta t \mathsf{G}(a_n)^2$ of descent: $\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \leq \Delta_n - \frac{1}{2} c_{\mathrm{KL}} \Delta t \mathsf{G}(a_n)^2 + B_{\mathrm{burn}}^{\mathrm{KL}} \Delta t^2$. Since $c_{\mathrm{KL}} \geq 1/9$ and Lemma 3.2 gives $\mathsf{G}(a_n)^2 \geq 2\alpha L_n^{\mathrm{KL}} \Delta_n$, $\frac{1}{2} c_{\mathrm{KL}} \Delta t \mathsf{G}(a_n)^2 \geq \frac{1}{16}\Delta t\, \alpha L_n^{\mathrm{KL}} \Delta_n$. This proves the stated one-step estimate. □

Theorem 10.5 (KL STL energy at covariance burn-in)**.** *Assume Assumptions 2.1 and 2.2 and $0 < \Delta t \leq \frac{1}{32\beta\lambda_{\max}}$. Define, for $0 \leq j \leq N$, $\Pi_{j,N}^{\mathrm{KL}} := \prod_{i=j}^{N-1}(1 - \frac{\alpha L_i^{\mathrm{KL}} \Delta t}{16})$, $\Pi_{N,N}^{\mathrm{KL}} := 1$. Then*

$$\boxed{\mathbb{E}\Delta_{N_{\mathrm{cov}}^{\mathrm{KL}}} \leq \Pi_{0,N_{\mathrm{cov}}^{\mathrm{KL}}}^{\mathrm{KL}} \Delta_0 + B_{\mathrm{burn}}^{\mathrm{KL}} \Delta t^2 \sum_{j=0}^{N_{\mathrm{cov}}^{\mathrm{KL}}-1} \Pi_{j+1,N_{\mathrm{cov}}^{\mathrm{KL}}}^{\mathrm{KL}}.} \tag{27}$$

In particular,

$$\boxed{\mathbb{E}\Delta_{N_{\mathrm{cov}}^{\mathrm{KL}}} \le \Delta_0 + 3\beta N_\theta^2 L_{H,\mathrm{g}}^2 \lambda_{\max}^4 \Delta t^2 N_{\mathrm{cov}}^{\mathrm{KL}}.} \tag{28}$$

Proof. By Lemma 10.4, for $0 \le n < N_{\mathrm{cov}}^{\mathrm{KL}}$, $\mathbb{E}\Delta_{n+1} \le (1 - \frac{\alpha L_n^{\mathrm{KL}} \Delta t}{16})\mathbb{E}\Delta_n + B_{\mathrm{burn}}^{\mathrm{KL}} \Delta t^2$. Iterating gives (27). Since every product factor is in $[0,1]$, the simple estimate (28) follows. $\square$

11 Gaussian target: global single-sample STL convergence with no floor

When the target is Gaussian, the STL covariance estimator is deterministic and the STL mean noise is multiplied by a factor that vanishes as the covariance reaches the target. This yields a global single-sample theorem with no asymptotic stochastic floor.

Throughout this section assume $\rho_{\mathrm{post}} = \mathcal{N}(m_\star, C_\star)$ and apply the affine change of variables $z = C_\star^{-1/2}(\theta - m_\star)$, after which $\rho_{\mathrm{post}} = \mathcal{N}(0, I)$ and $\nabla V(x) = x$, $\nabla^2 V(x) = I$. In these coordinates $\alpha = \beta = 1$, $\kappa = 1$, $\lambda_{\max} = \max\{\lambda_{0,\max}, 1\}$, and the energy gap is

$$\Delta_n = \tfrac{1}{2}\big(\|m_n\|^2 + \mathrm{Tr}(C_n) - \log\det C_n - N_\theta\big).$$

Lemma 11.1 (Energy equivalence on the band). *If $\frac{1}{2}I \preceq C \preceq 2I$ and $\rho_{\mathrm{post}} = \mathcal{N}(0, I)$, then*

$$\tfrac{1}{16}\big(\|m\|^2 + \|C - I\|_{\mathrm{F}}^2\big) \le \Delta(a) \le \|m\|^2 + \|C - I\|_{\mathrm{F}}^2.$$

Proof. Write $\Delta(a) = \frac{1}{2}\|m\|^2 + \frac{1}{2}\sum_i(\lambda_i - 1 - \log\lambda_i)$, where $\lambda_i \in [\frac{1}{2}, 2]$ are the eigenvalues of C. By Taylor's theorem with the second derivative $\frac{\mathrm{d}^2}{\mathrm{d}\lambda^2}(\lambda - 1 - \log\lambda) = \lambda^{-2}$, for $\lambda \in [\frac{1}{2}, 2]$ there is $\xi \in [\frac{1}{2}, 2]$ with $\lambda - 1 - \log\lambda = \frac{(\lambda-1)^2}{2\xi^2} \in [\frac{1}{8}(\lambda-1)^2, 2(\lambda-1)^2]$. Summing, $\frac{1}{16}\|C - I\|_{\mathrm{F}}^2 \le \frac{1}{2}\sum_i(\lambda_i - 1 - \log\lambda_i) \le \|C - I\|_{\mathrm{F}}^2$. Combining with $\frac{1}{2}\|m\|^2$ and using $\frac{1}{16} \le \frac{1}{2} \le 1$ gives the claim. $\square$

Theorem 11.2 (Gaussian target, no stochastic floor). *Assume $\rho_{\mathrm{post}} = \mathcal{N}(0, I)$ (after target whitening) and consider either the STL Riemannian scheme (7) or the STL KL scheme (8) with*

$$0 < \Delta t \le \min\left\{\tfrac{1}{2}, \tfrac{1}{\lambda_{\max}}\right\}.$$

Then the covariance evolves deterministically and reaches the band $\frac{1}{2}I \preceq C_n \preceq 2I$ for all $n \ge N_{\mathrm{cov}}$, where $N_{\mathrm{cov}} = O(\frac{1}{\Delta t}\log_+ \frac{1}{\lambda_{0,\min}} + \frac{1}{\Delta t}\log_+ \lambda_{0,\max})$. For all $n \ge N_{\mathrm{cov}}$,

$$\boxed{\mathbb{E}\Delta_n \le 320\Big(1 - \frac{\Delta t}{2}\Big)^{n - N_{\mathrm{cov}}} \mathbb{E}\Delta_{N_{\mathrm{cov}}}.}$$

In particular, neither scheme has an asymptotic stochastic noise floor. For the KL scheme the covariance convergence is exactly geometric: $C_n^{-1} - I = (1 + \Delta t)^{-n}(C_0^{-1} - I)$.

Proof. In whitened coordinates $H_n = \nabla^2 V(X_n) = I$, so the STL residual estimator $\widehat{K}_n = C_n^{-1} - I$ is deterministic and the covariance updates (9) are deterministic and autonomous.

Covariance, KL scheme. The update is $C_{n+1}^{-1} = \frac{1}{1+\Delta t}(C_n^{-1} + \Delta t I)$, so $C_{n+1}^{-1} - I = \frac{1}{1+\Delta t}(C_n^{-1} - I)$ and $C_n^{-1} - I = (1 + \Delta t)^{-n}(C_0^{-1} - I)$. Hence $C_n - I = -C_n(C_n^{-1} - I)$ gives, once $C_n \preceq 2I$, $\|C_n - I\|_{\mathrm{F}} \le 2(1 + \Delta t)^{-(n - N_{\mathrm{cov}})}\|C_{N_{\mathrm{cov}}} - I\|_{\mathrm{F}}$.

Covariance, Riemannian scheme. Since I commutes with C_n, the update is $C_{n+1} = C_n e^{\Delta t(I-C_n)}$, diagonal in the fixed eigenbasis of C_0, with eigenvalue map $\lambda_{i,n+1} = \lambda_{i,n} e^{\Delta t(1-\lambda_{i,n})}$. In log-coordinates $\mu_{i,n} = \log \lambda_{i,n}$, the map is $\mu \mapsto \mu + \Delta t(1 - e^{\mu})$ with derivative $1 - \Delta t e^{\mu}$. On the band $\mu \in [\log \frac{1}{2}, \log 2]$, i.e. $e^{\mu} \in [\frac{1}{2}, 2]$, and for $\Delta t \le \frac{1}{2}$,

$$|1 - \Delta t e^{\mu}| \le \max\{|1 - 2\Delta t|, |1 - \tfrac{\Delta t}{2}|\} = 1 - \tfrac{\Delta t}{2},$$

so $|\mu_{i,n+1}| \le (1 - \frac{\Delta t}{2})|\mu_{i,n}|$ once in the band. Since on the band $\frac{1}{2}|\lambda - 1| \le |\mu| = |\log \lambda| \le 2|\lambda - 1|$, we get $\|C_n - I\|_{\mathrm{F}} \le 2(1 - \frac{\Delta t}{2})^{n-N_{\mathrm{cov}}} \|C_{N_{\mathrm{cov}}} - I\|_{\mathrm{F}}$ (the factor 2 from $|\lambda - 1| \le e^{|\mu|}|\mu| \le 2|\mu|$). The two-sided pathwise bootstrap (Remark 9.5, here with $\alpha = \beta = 1$) gives the band after N_{cov} iterations. In both cases, with $\rho := e^{-\Delta t/2} \ge \max\{(1 + \Delta t)^{-1}, 1 - \frac{\Delta t}{2}\}$,

$$\|C_n - I\|_{\mathrm{F}}^2 \le 4\,\rho^{2(n-N_{\mathrm{cov}})}\,\Sigma_0, \qquad \Sigma_0 := \|C_{N_{\mathrm{cov}}} - I\|_{\mathrm{F}}^2\,. \tag{29}$$

Mean recursion. For both schemes the mean update is $m_{n+1} = m_n + \Delta t\, C_n \widehat{b}_n$ with, in whitened coordinates and $X_n = m_n + C_n^{1/2} Z_n$, $Z_n \sim \mathcal{N}(0, I)$,

$$\widehat{b}_n = -X_n + C_n^{-1}(X_n - m_n) = -m_n + (C_n^{-1} - I)C_n^{1/2} Z_n,$$

so $m_{n+1} = (I - \Delta t\, C_n)m_n + \Delta t(I - C_n)C_n^{1/2} Z_n$. Taking the conditional second moment, $\mathbb{E}[Z_n Z_n^{\top}] = I$ gives

$$\mathbb{E}[\|m_{n+1}\|^2 \mid \mathcal{F}_n] = \|(I - \Delta t\, C_n)m_n\|^2 + \Delta t^2 \operatorname{Tr}\big((I - C_n)C_n(I - C_n)\big).$$

On the band, the eigenvalues of $I - \Delta t\, C_n$ lie in $[1 - 2\Delta t, 1 - \frac{\Delta t}{2}] \subset [0, 1)$ (using $\Delta t \le \frac{1}{2}$), so $\|(I - \Delta t\, C_n)m_n\|^2 \le (1 - \frac{\Delta t}{2})^2 \|m_n\|^2 \le (1 - \frac{\Delta t}{2}) \|m_n\|^2$; and $\operatorname{Tr}((I - C_n)C_n(I - C_n)) \le \|C_n\|_{\mathrm{op}} \|C_n - I\|_{\mathrm{F}}^2 \le 2\,\|C_n - I\|_{\mathrm{F}}^2$. With (29),

$$\mathbb{E}[\|m_{n+1}\|^2 \mid \mathcal{F}_n] \le \big(1 - \tfrac{\Delta t}{2}\big) \|m_n\|^2 + 8\Delta t^2 \rho^{2(n-N_{\mathrm{cov}})} \Sigma_0.$$

Set $\rho_1 := 1 - \frac{\Delta t}{2}$ and $\rho_2 := \rho^2 = e^{-\Delta t}$. For $\Delta t \le \frac{1}{2}$ one has $\rho_2 \le \rho_1$ and $\rho_1 - \rho_2 \ge \frac{\Delta t}{4}$ (since $e^{-\Delta t} \le 1 - \Delta t + \frac{\Delta t^2}{2}$ and $(1 - \frac{\Delta t}{2}) - (1 - \Delta t + \frac{\Delta t^2}{2}) = \frac{\Delta t}{2}(1 - \Delta t) \ge \frac{\Delta t}{4}$). Unrolling the linear recursion from N_{cov},

$$\mathbb{E}\,\|m_n\|^2 \le \rho_1^{n-N_{\mathrm{cov}}} \mathbb{E}\,\|m_{N_{\mathrm{cov}}}\|^2 + 8\Delta t^2 \Sigma_0 \sum_{k=0}^{n-N_{\mathrm{cov}}-1} \rho_1^{n-N_{\mathrm{cov}}-1-k} \rho_2^k \le \rho_1^{n-N_{\mathrm{cov}}} \Big(\mathbb{E}\,\|m_{N_{\mathrm{cov}}}\|^2 + \tfrac{8\Delta t^2 \Sigma_0}{\rho_1 - \rho_2}\Big).$$

Since $\frac{8\Delta t^2}{\rho_1 - \rho_2} \le \frac{8\Delta t^2}{\Delta t/4} = 32\Delta t \le 16$, $\mathbb{E}\,\|m_n\|^2 \le \rho_1^{n-N_{\mathrm{cov}}}(\mathbb{E}\,\|m_{N_{\mathrm{cov}}}\|^2 + 16\,\mathbb{E}\Sigma_0)$.

Assembly. Combining with (29) (and $\rho^2 \le \rho_1$),

$$\mathbb{E}\big[\|m_n\|^2 + \|C_n - I\|_{\mathrm{F}}^2\big] \le \rho_1^{n-N_{\mathrm{cov}}} \big(\mathbb{E}\,\|m_{N_{\mathrm{cov}}}\|^2 + 16\,\mathbb{E}\Sigma_0 + 4\,\mathbb{E}\Sigma_0\big) \le 20\,\rho_1^{n-N_{\mathrm{cov}}}\,\mathbb{E}\big[\|m_{N_{\mathrm{cov}}}\|^2 + \Sigma_0\big].$$

By Lemma 11.1, $\mathbb{E}\Delta_n \le \mathbb{E}[\|m_n\|^2 + \|C_n - I\|_{\mathrm{F}}^2]$ and $\mathbb{E}[\|m_{N_{\mathrm{cov}}}\|^2 + \Sigma_0] \le 16\,\mathbb{E}\Delta_{N_{\mathrm{cov}}}$, so $\mathbb{E}\Delta_n \le 20 \cdot 16\,\rho_1^{n-N_{\mathrm{cov}}} \mathbb{E}\Delta_{N_{\mathrm{cov}}} = 320(1 - \frac{\Delta t}{2})^{n-N_{\mathrm{cov}}} \mathbb{E}\Delta_{N_{\mathrm{cov}}}$. □

12 General target: global stochastic STL rates

We now treat a general α-strongly-log-concave, β-smooth target. After two-sided burn-in the iterates lie in the band $\frac{1}{2\beta} I \preceq C_n \preceq \frac{2}{\alpha} I$ (Remark 9.5), on which we run the stochastic descent.

12.1 Riemannian STL scheme

Lemma 12.1 (Riemannian STL one-step descent)**.** *Under Assumption 2.1, suppose a_n lies in the band $\frac{1}{2\beta}I \preceq C_n \preceq \frac{2}{\alpha}I$ and $0 < \Delta t \leq \frac{1}{4\kappa}$. Then the STL Riemannian scheme (7) satisfies*

$$\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \leq \Big(1 - \tfrac{\Delta t}{2\kappa}\Big)\Delta_n + 2\kappa\Delta t^2\Psi(a_n).$$

Proof. By Proposition 12.2 below, the stochastic update is the geodesic step $a_{n+1} = \mathrm{Exp}_{a_n}(-\Delta t\,\widehat{\mathrm{grad}}\mathcal{E}(a_n))$, whose covariance eigenvalues move monotonically from those of C_n to those of C_{n+1}, so the geodesic stays in the band. The geodesic L-smoothness of $\mathcal{E}$ on the band, with $L = \beta \cdot \frac{2}{\alpha} = 2\kappa$ (the base-manuscript constant $L = \beta\Lambda$ with covariance bound $\Lambda = \frac{2}{\alpha}$), gives

$$\mathcal{E}(a_{n+1}) \leq \mathcal{E}(a_n) + \Big\langle \mathrm{grad}\,\mathcal{E}(a_n), -\Delta t\,\widehat{\mathrm{grad}}\mathcal{E}(a_n)\Big\rangle_{a_n} + \frac{L\Delta t^2}{2}\Big\|\widehat{\mathrm{grad}}\mathcal{E}(a_n)\Big\|_{a_n}^2.$$

Write $\xi_n := \widehat{\mathrm{grad}}\mathcal{E}(a_n) - \mathrm{grad}\,\mathcal{E}(a_n) = \xi(a_n; X_n)$. Subtracting $\mathcal{E}(a_\star)$, taking $\mathbb{E}[\cdot \mid \mathcal{F}_n]$, and using $\mathbb{E}[\xi_n \mid \mathcal{F}_n] = 0$,

$$\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \leq \Delta_n - \Delta t\Big(1 - \tfrac{L\Delta t}{2}\Big)\|\mathrm{grad}\,\mathcal{E}(a_n)\|_{a_n}^2 + \tfrac{L\Delta t^2}{2}\,\mathbb{E}[\|\xi_n\|_{a_n}^2 \mid \mathcal{F}_n].$$

With the convention (10), $\|\mathrm{grad}\,\mathcal{E}(a_n)\|_{a_n}^2 = \mathsf{G}(a_n)^2$, and by Lemma 9.1, $\mathbb{E}[\|\xi_n\|_{a_n}^2 \mid \mathcal{F}_n] \leq \mathsf{G}(a_n)^2 + 2\Psi(a_n)$. Hence

$$\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \leq \Delta_n - \Delta t(1 - L\Delta t)\mathsf{G}(a_n)^2 + L\Delta t^2\Psi(a_n).$$

Since $\Delta t \leq \frac{1}{4\kappa}$ gives $L\Delta t = 2\kappa\Delta t \leq \frac{1}{2}$, the coefficient $1 - L\Delta t \geq \frac{1}{2}$. By Lemma 3.2 with $\ell = \frac{1}{2\beta}$, $\mathsf{G}(a_n)^2 \geq \frac{1}{2\beta} \cdot 2\alpha\Delta_n = \frac{1}{\kappa}\Delta_n$. Therefore

$$\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \leq \Delta_n - \tfrac{\Delta t}{2}\mathsf{G}(a_n)^2 + 2\kappa\Delta t^2\Psi(a_n) \leq \Big(1 - \tfrac{\Delta t}{2\kappa}\Big)\Delta_n + 2\kappa\Delta t^2\Psi(a_n). \qquad \square$$

Proposition 12.2 (Geodesic form of the stochastic step)**.** *The STL Riemannian update (7) is $a_{n+1} = \mathrm{Exp}_{a_n}(-\Delta t\,\widehat{\mathrm{grad}}\mathcal{E}(a_n))$ with $\widehat{\mathrm{grad}}\mathcal{E}(a_n) = (-C_n\widehat{b}_n, -C_n\widehat{K}_nC_n)$, where $\mathrm{Exp}_{(m,C)}(u, U) = (m + u,\, C^{1/2}\exp(C^{-1/2}UC^{-1/2})C^{1/2})$.*

Proof. The mean component is immediate. For the covariance, with $U = \Delta t\, C_n\widehat{K}_nC_n$, $C_n^{-1/2}(-\Delta t\,(-C_n\widehat{K}_nC_n))C_n^{-}$ $\Delta t\, C_n^{1/2}\widehat{K}_nC_n^{1/2}$, so the exponential-map covariance component equals $C_n^{1/2}\exp(\Delta t\, C_n^{1/2}\widehat{K}_nC_n^{1/2})C_n^{1/2}$, which is (7). $\square$

Theorem 12.3 (Riemannian STL global stochastic rate)**.** *Under Assumption 2.1, suppose*

$$0 < \Delta t \leq \min\left\{1, \frac{1}{\beta\lambda_{\max}}, \frac{1}{4\kappa}\right\}.$$

Then the STL Riemannian scheme (7) satisfies, for all $N \geq N_{\mathrm{cov}}^{\mathrm{R}}$,

$$\boxed{\mathbb{E}\Delta_N \leq \Big(1 - \tfrac{\Delta t}{2\kappa}\Big)^{N - N_{\mathrm{cov}}^{\mathrm{R}}}\mathbb{E}\Delta_{N_{\mathrm{cov}}^{\mathrm{R}}} + 4\kappa^2\Delta t\,\Psi_\star^{\mathrm{R}}.}$$

In particular, using $\Psi_\star^{\mathrm{R}} \leq 4N_\theta(\kappa - 1)^2$ (Lemma 9.3),

$$\mathbb{E}\Delta_N \leq \Big(1 - \tfrac{\Delta t}{2\kappa}\Big)^{N - N_{\mathrm{cov}}^{\mathrm{R}}}\mathbb{E}\Delta_{N_{\mathrm{cov}}^{\mathrm{R}}} + 16\kappa^2(\kappa - 1)^2N_\theta\,\Delta t,$$

and $\limsup_{N\to\infty}\mathbb{E}\Delta_N \leq 16\kappa^2(\kappa - 1)^2N_\theta\,\Delta t$.

Proof. By Remark 9.5, $\frac{1}{2\beta}I \preceq C_n \preceq \frac{2}{\alpha}I$ for all $n \ge N^{\mathrm{R}}_{\mathrm{cov}}$. Lemma 12.1 and $\Psi(a_n) \le \Psi^{\mathrm{R}}_\star$ give, for $n \ge N^{\mathrm{R}}_{\mathrm{cov}}$, $\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \le (1 - \frac{\Delta t}{2\kappa})\Delta_n + 2\kappa\Delta t^2\Psi^{\mathrm{R}}_\star$. Taking total expectations and unrolling from $N^{\mathrm{R}}_{\mathrm{cov}}$, the geometric series sums to

$$\mathbb{E}\Delta_N \le \left(1 - \tfrac{\Delta t}{2\kappa}\right)^{N-N^{\mathrm{R}}_{\mathrm{cov}}}\mathbb{E}\Delta_{N^{\mathrm{R}}_{\mathrm{cov}}} + 2\kappa\Delta t^2\Psi^{\mathrm{R}}_\star \cdot \frac{1}{\Delta t/(2\kappa)} = \left(1 - \tfrac{\Delta t}{2\kappa}\right)^{N-N^{\mathrm{R}}_{\mathrm{cov}}}\mathbb{E}\Delta_{N^{\mathrm{R}}_{\mathrm{cov}}} + 4\kappa^2\Delta t\,\Psi^{\mathrm{R}}_\star.$$

The crude form and the lim sup follow from Lemma 9.3. □

Corollary 12.4 (Closed Riemannian STL global rate including burn-in)**.** *Assume Assumptions 2.1 and 2.2 and* $0 < \Delta t \le \min\{\frac{1}{2\beta\lambda_{\max}}, \frac{1}{4\kappa}\}$. *Then, for all* $N \ge N^{\mathrm{R}}_{\mathrm{cov}}$*,*

$$\boxed{\mathbb{E}\Delta_N \le \left(1 - \tfrac{\Delta t}{2\kappa}\right)^{N-N^{\mathrm{R}}_{\mathrm{cov}}}\mathcal{B}^{\mathrm{R}}_{\mathrm{burn}} + 4\kappa^2\Delta t\,\Psi^{\mathrm{R}}_\star,}$$

where the fully explicit burn-in energy bound is $\mathcal{B}^{\mathrm{R}}_{\mathrm{burn}} := \Pi^{\mathrm{R}}_{0,N^{\mathrm{R}}_{\mathrm{cov}}}\Delta_0 + B^{\mathrm{R}}_{\mathrm{burn}}\Delta t^2\sum_{j=0}^{N^{\mathrm{R}}_{\mathrm{cov}}-1}\Pi^{\mathrm{R}}_{j+1,N^{\mathrm{R}}_{\mathrm{cov}}}$. *In particular,*

$$\boxed{\mathbb{E}\Delta_N \le \left(1 - \tfrac{\Delta t}{2\kappa}\right)^{N-N^{\mathrm{R}}_{\mathrm{cov}}}\left(\Delta_0 + \beta N_\theta^2 L_{H,\mathrm{g}}^2\lambda_{\max}^4\Delta t^2 N^{\mathrm{R}}_{\mathrm{cov}}\right) + 4\kappa^2\Delta t\,\Psi^{\mathrm{R}}_\star.}$$

Using $\Psi^{\mathrm{R}}_\star \le 4N_\theta(\kappa-1)^2$ *further gives the explicit floor* $16\kappa^2(\kappa-1)^2N_\theta\Delta t$*.*

Proof. Combine the post-burn-in estimate of Theorem 12.3 with Theorem 10.2, which bounds $\mathbb{E}\Delta_{N^{\mathrm{R}}_{\mathrm{cov}}}$ by $\mathcal{B}^{\mathrm{R}}_{\mathrm{burn}}$. The simplified version uses (25), and the final floor uses Lemma 9.3. □

12.2 KL STL scheme

For the KL scheme we compare the stochastic update a_{n+1} to the deterministic update $\bar{a}_{n+1}$ (the right-hand side of (3) evaluated at a_n). Recall $D(a, a') = \mathrm{KL}(\rho_a \,\|\, \rho_{a'})$, and write $\bar{P} := \bar{C}_{n+1}^{-1}$, $P := C_{n+1}^{-1}$,

$$W := P - \bar{P} = -\frac{\Delta t}{1+\Delta t}\,\varepsilon_R, \qquad Y_n := \bar{C}_{n+1}^{1/2}W\bar{C}_{n+1}^{1/2},$$

so that $\mathbb{E}[W \mid \mathcal{F}_n] = 0$ and $\bar{C}_{n+1}^{-1/2}C_{n+1}\bar{C}_{n+1}^{-1/2} = (I + Y_n)^{-1}$.

Lemma 12.5 (KL STL covariance perturbation)**.** *Under Assumption 2.1, if* a_n *lies in the band* $\frac{1}{2\beta}I \preceq C_n \preceq \frac{2}{\alpha}I$ *and* $0 < \Delta t \le \frac{1}{4\kappa}$*, then* $a_{n+1}, \bar{a}_{n+1}$ *also lie in the band,* $\|Y_n\|_{\mathrm{op}} \le \frac{1}{2}$*, and*

$$\mathbb{E}[\|Y_n\|_{\mathrm{F}}^2 \mid \mathcal{F}_n] \le 16\kappa^2\Delta t^2\,\Psi(a_n), \qquad \mathbb{E}[\varepsilon_b^\top C_n\varepsilon_b \mid \mathcal{F}_n] \le \mathsf{G}(a_n)^2 + \Psi(a_n).$$

Proof. The band membership of $a_{n+1}, \bar{a}_{n+1}$ is the pathwise bootstrap (Remark 9.5). Since $\|\varepsilon_R\|_{\mathrm{op}} \le \beta - \alpha$ and $\frac{\Delta t}{1+\Delta t} \le \Delta t$, $\|W\|_{\mathrm{op}} \le \Delta t(\beta-\alpha)$, hence $\|Y_n\|_{\mathrm{op}} \le \left\|\bar{C}_{n+1}\right\|_{\mathrm{op}}\|W\|_{\mathrm{op}} \le \frac{2}{\alpha}\Delta t(\beta-\alpha) = 2\Delta t(\kappa-1) \le \frac{\kappa-1}{2\kappa} \le \frac{1}{2}$, using $\Delta t \le \frac{1}{4\kappa}$. For the Frobenius bound, $\mathbb{E}[\|Y_n\|_{\mathrm{F}}^2 \mid \mathcal{F}_n] \le \left\|\bar{C}_{n+1}\right\|_{\mathrm{op}}^2\mathbb{E}[\|W\|_{\mathrm{F}}^2 \mid \mathcal{F}_n] \le \frac{4}{\alpha^2}\Delta t^2\mathbb{E}[\|\varepsilon_R\|_{\mathrm{F}}^2 \mid \mathcal{F}_n]$, and since $\Psi(a_n) = \mathbb{E}[\left\|C_n^{1/2}\varepsilon_R C_n^{1/2}\right\|_{\mathrm{F}}^2 \mid \mathcal{F}_n] \ge \lambda_{\min}(C_n)^2\mathbb{E}[\|\varepsilon_R\|_{\mathrm{F}}^2 \mid \mathcal{F}_n] \ge \frac{1}{4\beta^2}\mathbb{E}[\|\varepsilon_R\|_{\mathrm{F}}^2 \mid \mathcal{F}_n]$, we get $\mathbb{E}[\|\varepsilon_R\|_{\mathrm{F}}^2 \mid \mathcal{F}_n] \le 4\beta^2\Psi(a_n)$ and hence $\mathbb{E}[\|Y_n\|_{\mathrm{F}}^2 \mid \mathcal{F}_n] \le \frac{4}{\alpha^2}\Delta t^2 \cdot 4\beta^2\Psi(a_n) = 16\kappa^2\Delta t^2\Psi(a_n)$. The last bound is (20) together with $\left\|C_n^{1/2}R(a_n)C_n^{1/2}\right\|_{\mathrm{F}}^2 \le \mathsf{G}(a_n)^2$. □

Lemma 12.6 (KL STL one-step descent)**.** *Under Assumption 2.1, if a_n lies in the band $\frac{1}{2\beta}I \preceq C_n \preceq \frac{2}{\alpha}I$ and $0 < \Delta t \le \frac{1}{288\kappa^3}$, then the STL KL scheme (8) satisfies*

$$\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \le \Big(1 - \tfrac{\Delta t}{18\kappa}\Big)\Delta_n + 68\kappa^3 \Delta t^2 \Psi(a_n).$$

Proof. Throughout, all conditional expectations are given $\mathcal{F}_n$, and we use $\Delta t \le \frac{1}{288\kappa^3} \le \frac{1}{4\kappa} \le \frac{1}{2\kappa} \le \frac{1}{2}$.

Step 1: entropy term. $\mathcal{E}_1 = \frac{1}{2}\log\det C^{-1} + \text{const.}$, and $\mathbb{E}[C_{n+1}^{-1} \mid \mathcal{F}_n] = \frac{1}{1+\Delta t}(C_n^{-1} + \Delta t\, A(a_n)) = \bar{C}_{n+1}^{-1}$. By concavity of $P \mapsto \log\det P$ and Jensen, $\mathbb{E}[\log\det C_{n+1}^{-1}] \le \log\det \bar{C}_{n+1}^{-1}$, hence $\mathbb{E}[\mathcal{E}_1(a_{n+1}) - \mathcal{E}_1(\bar{a}_{n+1})] \le 0$.

Step 2: potential term via relative smoothness. By the relative smoothness of $\mathcal{E}_2$ on the band (base-manuscript constant $L = \beta \cdot \frac{2}{\alpha} = 2\kappa$),

$$\mathcal{E}_2(a_{n+1}) \le \mathcal{E}_2(\bar{a}_{n+1}) + \langle \nabla \mathcal{E}_2(\bar{a}_{n+1}), a_{n+1} - \bar{a}_{n+1}\rangle + 2\kappa\, D(a_{n+1}, \bar{a}_{n+1}).$$

The mean part of the inner product is conditionally centered, since $m_{n+1} - \bar{m}_{n+1} = \Delta t\, C_n \varepsilon_b$ and $\mathbb{E}[\varepsilon_b \mid \mathcal{F}_n] = 0$. For the covariance part, $\nabla_C \mathcal{E}_2(\bar{a}_{n+1}) = \frac{1}{2}A(\bar{a}_{n+1})$ and

$$C_{n+1} - \bar{C}_{n+1} = \bar{C}_{n+1}^{1/2}\big((I+Y_n)^{-1} - I\big)\bar{C}_{n+1}^{1/2} = \bar{C}_{n+1}^{1/2}\big(-Y_n + Y_n^2(I+Y_n)^{-1}\big)\bar{C}_{n+1}^{1/2}.$$

Taking $\mathbb{E}[\,\cdot \mid \mathcal{F}_n]$ and using $\mathbb{E}[Y_n \mid \mathcal{F}_n] = 0$,

$$\mathbb{E}\big[\big\langle \nabla_C \mathcal{E}_2(\bar{a}_{n+1}), C_{n+1} - \bar{C}_{n+1}\big\rangle\big] = \tfrac{1}{2}\mathbb{E}\big[\mathrm{Tr}\big(\widetilde{A}_n\, Y_n^2 (I+Y_n)^{-1}\big)\big], \quad \widetilde{A}_n := \bar{C}_{n+1}^{1/2} A(\bar{a}_{n+1}) \bar{C}_{n+1}^{1/2},$$

where $\widetilde{A}_n \succeq 0$ with $\big\|\widetilde{A}_n\big\|_{\mathrm{op}} \le \beta \cdot \frac{2}{\alpha} = 2\kappa$. The matrix $Y_n^2(I+Y_n)^{-1}$ is symmetric with eigenvalues $y_i^2/(1+y_i)$, $|y_i| \le \frac{1}{2}$, so $\mathrm{Tr}\,|Y_n^2(I+Y_n)^{-1}| \le 2\,\|Y_n\|_{\mathrm{F}}^2$. Hence

$$\big|\mathbb{E}[\big\langle \nabla_C \mathcal{E}_2(\bar{a}_{n+1}), C_{n+1} - \bar{C}_{n+1}\big\rangle]\big| \le \tfrac{1}{2} \cdot 2\kappa \cdot 2\,\mathbb{E}\,\|Y_n\|_{\mathrm{F}}^2 = 2\kappa\,\mathbb{E}\,\|Y_n\|_{\mathrm{F}}^2 \le 32\kappa^3 \Delta t^2 \Psi(a_n),$$

using Lemma 12.5.

Step 3: the divergence $D(a_{n+1}, \bar{a}_{n+1})$. The mean part is $D^m = \frac{\Delta t^2}{2}\varepsilon_b^\top C_n \bar{C}_{n+1}^{-1} C_n \varepsilon_b$; since $C_n \bar{C}_{n+1}^{-1} C_n \preceq \big\|\bar{C}_{n+1}^{-1}\big\|_{\mathrm{op}} \|C_n\|_{\mathrm{op}}\, C_n \preceq 2\beta \cdot \frac{2}{\alpha}\, C_n = 4\kappa C_n$,

$$\mathbb{E}[D^m] \le 2\kappa \Delta t^2\, \mathbb{E}[\varepsilon_b^\top C_n \varepsilon_b] \le 2\kappa \Delta t^2 (\mathsf{G}(a_n)^2 + \Psi(a_n)).$$

The covariance part is $D^C = \frac{1}{2}\,\mathrm{Tr}((I+Y_n)^{-1} + \log(I+Y_n) - I)$; the scalar inequality $\frac{1}{1+y} + \log(1+y) - 1 \le \frac{y^2}{2(1-\rho)^2}$ for $|y| \le \rho \le \frac{1}{2}$ gives, with $\rho = \|Y_n\|_{\mathrm{op}} \le \frac{1}{2}$, $D^C \le \|Y_n\|_{\mathrm{F}}^2$, so $\mathbb{E}[D^C] \le 16\kappa^2 \Delta t^2 \Psi(a_n)$. Therefore

$$\mathbb{E}[D(a_{n+1}, \bar{a}_{n+1})] \le 2\kappa \Delta t^2 \mathsf{G}(a_n)^2 + (2\kappa + 16\kappa^2)\Delta t^2 \Psi(a_n) \le 2\kappa \Delta t^2 \mathsf{G}(a_n)^2 + 18\kappa^2 \Delta t^2 \Psi(a_n).$$

Step 4: collecting the stochastic excess. Combining Steps 1–3,

$$\mathbb{E}[\mathcal{E}(a_{n+1}) - \mathcal{E}(\bar{a}_{n+1})] \le 32\kappa^3 \Delta t^2 \Psi + 2\kappa\big(2\kappa \Delta t^2 \mathsf{G}^2 + 18\kappa^2 \Delta t^2 \Psi\big) = 4\kappa^2 \Delta t^2 \mathsf{G}(a_n)^2 + 68\kappa^3 \Delta t^2 \Psi(a_n),$$

writing $\mathsf{G} = \mathsf{G}(a_n)$, $\Psi = \Psi(a_n)$.

Step 5: deterministic descent with a gradient margin. The base-manuscript KL one-step estimate is $\Delta(\bar{a}_{n+1}) \le \Delta_n - \frac{1}{\Delta t} D(a_n, \bar{a}_{n+1})$ with, on the band ($\lambda_{\min}(C_n) \ge \frac{1}{2\beta}$, $\lambda_{\max} = \frac{2}{\alpha}$),

$$D(a_n, \bar{a}_{n+1}) \ge \frac{\lambda_{\min}(C_n)\Delta t^2}{4(1+\Delta t)(1+\Delta t \beta \lambda_{\max})} \left\|\mathrm{grad}^{W_2}\, \mathcal{E}(a_n)\right\|_{W_2}^2 \ge \frac{\Delta t^2}{18\beta} \left\|\mathrm{grad}^{W_2}\, \mathcal{E}(a_n)\right\|_{W_2}^2,$$

using $(1+\Delta t)(1+2\kappa\Delta t) \le \frac{9}{4}$ (as $\Delta t \le \frac{1}{2\kappa}$). By Lemma 3.1, $\left\|\operatorname{grad}^{W_2} \mathcal{E}(a_n)\right\|^2_{W_2} \ge 2\alpha\Delta_n$, and by the upper bound in Lemma 3.2, $\left\|\operatorname{grad}^{W_2} \mathcal{E}(a_n)\right\|^2_{W_2} \ge \frac{\alpha}{2}\mathsf{G}(a_n)^2$. Hence $\frac{1}{\Delta t}D(a_n, \bar a_{n+1}) \ge \max\{\frac{\Delta t}{9\kappa}\Delta_n, \frac{\Delta t}{36\kappa}\mathsf{G}^2\} \ge \frac{\Delta t}{18\kappa}\Delta_n + \frac{\Delta t}{72\kappa}\mathsf{G}^2$, so that $\Delta(\bar a_{n+1}) \le (1-\frac{\Delta t}{18\kappa})\Delta_n - \frac{\Delta t}{72\kappa}\mathsf{G}(a_n)^2$.

Step 6: combine. Adding the excess of Step 4 to the descent of Step 5,

$$\mathbb{E}[\Delta_{n+1}] \le \left(1 - \tfrac{\Delta t}{18\kappa}\right)\Delta_n - \tfrac{\Delta t}{72\kappa}\mathsf{G}(a_n)^2 + 4\kappa^2\Delta t^2\mathsf{G}(a_n)^2 + 68\kappa^3\Delta t^2\Psi(a_n).$$

Since $\Delta t \le \frac{1}{288\kappa^3}$ gives $4\kappa^2\Delta t^2 \le \frac{\Delta t}{72\kappa}$, the two G^2 terms cancel (in sign), yielding the claim. □

Theorem 12.7 (KL STL global stochastic rate)**.** *Under Assumption 2.1, suppose* $0 < \Delta t \le \min\{\frac{1}{\beta\lambda_{\max}}, \frac{1}{288\kappa^3}\}$*. Then the STL KL scheme* (8) *satisfies, for all* $N \ge N^{\mathrm{KL}}_{\mathrm{cov}}$*,*

$$\boxed{\mathbb{E}\Delta_N \le \left(1 - \tfrac{\Delta t}{18\kappa}\right)^{N-N^{\mathrm{KL}}_{\mathrm{cov}}}\mathbb{E}\Delta_{N^{\mathrm{KL}}_{\mathrm{cov}}} + 1224\kappa^4\Delta t\,\Psi^{\mathrm{KL}}_\star.}$$

In particular, using $\Psi^{\mathrm{KL}}_\star \le 4N_\theta(\kappa-1)^2$*,*

$$\mathbb{E}\Delta_N \le \left(1 - \tfrac{\Delta t}{18\kappa}\right)^{N-N^{\mathrm{KL}}_{\mathrm{cov}}}\mathbb{E}\Delta_{N^{\mathrm{KL}}_{\mathrm{cov}}} + 4896\kappa^4(\kappa-1)^2N_\theta\,\Delta t,$$

and $\limsup_{N\to\infty}\mathbb{E}\Delta_N \le 4896\kappa^4(\kappa-1)^2N_\theta\,\Delta t$*.*

Proof. By Remark 9.5, the band holds for all $n \ge N^{\mathrm{KL}}_{\mathrm{cov}}$. Lemma 12.6 and $\Psi(a_n) \le \Psi^{\mathrm{KL}}_\star$ give, for $n \ge N^{\mathrm{KL}}_{\mathrm{cov}}$, $\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] \le (1-\frac{\Delta t}{18\kappa})\Delta_n + 68\kappa^3\Delta t^2\Psi^{\mathrm{KL}}_\star$. Unrolling, the floor is $68\kappa^3\Delta t^2\Psi^{\mathrm{KL}}_\star \cdot \frac{18\kappa}{\Delta t} = 1224\kappa^4\Delta t\Psi^{\mathrm{KL}}_\star$; the crude form follows from Lemma 9.3. □

Corollary 12.8 (Closed KL STL global rate including burn-in)**.** *Assume Assumptions 2.1 and 2.2 and* $0 < \Delta t \le \min\{\frac{1}{32\beta\lambda_{\max}}, \frac{1}{288\kappa^3}\}$*. Then, for all* $N \ge N^{\mathrm{KL}}_{\mathrm{cov}}$*,*

$$\boxed{\mathbb{E}\Delta_N \le \left(1 - \tfrac{\Delta t}{18\kappa}\right)^{N-N^{\mathrm{KL}}_{\mathrm{cov}}}\mathcal{B}^{\mathrm{KL}}_{\mathrm{burn}} + 1224\kappa^4\Delta t\,\Psi^{\mathrm{KL}}_\star,}$$

where $\mathcal{B}^{\mathrm{KL}}_{\mathrm{burn}} := \Pi^{\mathrm{KL}}_{0,N^{\mathrm{KL}}_{\mathrm{cov}}}\Delta_0 + B^{\mathrm{KL}}_{\mathrm{burn}}\Delta t^2\sum_{j=0}^{N^{\mathrm{KL}}_{\mathrm{cov}}-1}\Pi^{\mathrm{KL}}_{j+1,N^{\mathrm{KL}}_{\mathrm{cov}}}$*. In particular,*

$$\boxed{\mathbb{E}\Delta_N \le \left(1 - \tfrac{\Delta t}{18\kappa}\right)^{N-N^{\mathrm{KL}}_{\mathrm{cov}}}\left(\Delta_0 + 3\beta N_\theta^2 L^2_{H,\mathrm{g}}\lambda^4_{\max}\Delta t^2 N^{\mathrm{KL}}_{\mathrm{cov}}\right) + 1224\kappa^4\Delta t\,\Psi^{\mathrm{KL}}_\star.}$$

Using $\Psi^{\mathrm{KL}}_\star \le 4N_\theta(\kappa-1)^2$ *further gives the explicit floor* $4896\kappa^4(\kappa-1)^2N_\theta\Delta t$*.*

Proof. Combine Theorem 12.7 with Theorem 10.5. The simplified version uses (28), and the final floor uses Lemma 9.3. □

Remark 12.9 (On the KL constants). The constants 1224, 4896 and the powers κ^4 (floor) and κ^{-3} (stepsize) in Theorem 12.7 are honest but *not optimized*. They arise because the covariance update of the KL scheme is a matrix-inverse perturbation: the natural object $Y_n = \bar C^{1/2}_{n+1}W\bar C^{1/2}_{n+1}$ is measured at $\bar C_{n+1}$ rather than at C_n, and converting between the two intrinsic scales (Lemma 12.5) and passing through relative smoothness each cost a factor of κ. The Riemannian scheme avoids the inverse and yields the cleaner κ^2 floor of Theorem 12.3. The qualitative conclusions—a floor proportional to $\Delta t\,\Psi^{\mathrm{KL}}_\star$, hence *zero* for a Gaussian target (where $\Psi \equiv 0$, recovering Theorem 11.2), and otherwise controlled by the condition number rather than by raw powers of $\beta, \alpha, \lambda_{\max}$—do not depend on optimizing these constants.

13 Stochastic global-to-local complexity

The global stochastic estimates of Theorems 12.3 and 12.7 are energy estimates with a κ-controlled floor. To reach the sharper local floor we chain them to the local STL theorems of the base manuscript, which require the whitened Hessian to be Lipschitz.

Assumption 13.1 (Whitened-Hessian Lipschitzness)**.** The whitened Hessian is L_H-Lipschitz on the relevant local region: $\left\|\nabla^2\widehat{V}(x) - \nabla^2\widehat{V}(y)\right\|_{\mathrm{op}} \le L_H \|x-y\|$, where $\widehat{V}(z) = V(m_\star + C_\star^{1/2} z)$.

Assumption 13.2 (Local STL theorems, original coordinates)**.** Under Assumptions 2.1 and 13.1, with $r_{\mathrm{R}}, r_{\mathrm{KL}}, \Gamma$ as in Assumption 7.1, the base manuscript provides the following. If the STL Riemannian iterates remain in $\{a : \|a - a_\star\|_\star \le r_{\mathrm{R}}\}$ and $0 < \Delta t \le \frac{1}{256(4+\Gamma)(48+12N_\theta L_H^2)}$, then

$$\mathbb{E}\,\|a_N - a_\star\|_\star^2 \le \left(1 - \tfrac{\Delta t}{2(4+\Gamma)}\right)^N \|a_0 - a_\star\|_\star^2 + 672\,\Delta t(4+\Gamma)N_\theta^2 L_H^2.$$

If the STL KL iterates remain in $\{a : \|a - a_\star\|_{\star,\Delta t} \le r_{\mathrm{KL}}\}$ and $0 < \Delta t \le \frac{1}{272(6+\Gamma)(48+12N_\theta L_H^2)}$, then

$$\mathbb{E}\,\|a_N - a_\star\|_{\star,\Delta t}^2 \le \left(1 - \tfrac{\Delta t}{2(4+2\Delta t+\Gamma)}\right)^N \|a_0 - a_\star\|_{\star,\Delta t}^2 + 714\,\Delta t(4+2\Delta t+\Gamma)N_\theta^2 L_H^2.$$

Theorem 13.3 (Riemannian STL global-to-local)**.** *Assume Assumptions 2.1, 13.1 and 13.2, the energy-to-local conversion of Remark 9.2 (i.e. $\|a - a_\star\|_\star^2 \le \frac{2\chi_\star A_0}{\alpha}\Delta(a)$ with threshold Δ_{R} from (14)), and that the STL Riemannian iterates remain in $\{a : \|a - a_\star\|_\star \le r_{\mathrm{R}}\}$ once they enter it. Suppose*

$$0 < \Delta t \le \min\left\{1, \frac{1}{\beta\lambda_{\max}}, \frac{1}{4\kappa}, \frac{1}{256(4+\Gamma)(48+12N_\theta L_H^2)}\right\} \quad \text{and} \quad 4\kappa^2\Delta t\,\Psi_\star^{\mathrm{R}} \le \tfrac{1}{2}\Delta_{\mathrm{R}}.$$

Then $\mathbb{E}\Delta_N \le \Delta_{\mathrm{R}}$ after

$$N_{\mathrm{entry}}^{\mathrm{R}} = N_{\mathrm{cov}}^{\mathrm{R}} + \left\lceil \frac{\log\left(2\,\mathbb{E}\Delta_{N_{\mathrm{cov}}^{\mathrm{R}}}/\Delta_{\mathrm{R}}\right)}{-\log(1 - \Delta t/(2\kappa))} \right\rceil$$

iterations, and a further $N_{\mathrm{loc}}^{\mathrm{R}}(\epsilon) = \lceil \log(r_{\mathrm{R}}^2/\epsilon)/(-\log(1-\Delta t/[2(4+\Gamma)])) \rceil$ iterations guarantee

$$\mathbb{E}\,\|a_N - a_\star\|_\star^2 \le \epsilon + 672\,\Delta t(4+\Gamma)N_\theta^2 L_H^2.$$

Proof. By Theorem 12.3, for $N \ge N_{\mathrm{cov}}^{\mathrm{R}}$, $\mathbb{E}\Delta_N \le (1 - \frac{\Delta t}{2\kappa})^{N - N_{\mathrm{cov}}^{\mathrm{R}}}\mathbb{E}\Delta_{N_{\mathrm{cov}}^{\mathrm{R}}} + 4\kappa^2\Delta t\Psi_\star^{\mathrm{R}}$. The floor condition makes the additive term $\le \frac{1}{2}\Delta_{\mathrm{R}}$, and the displayed entry time makes the geometric term $\le \frac{1}{2}\Delta_{\mathrm{R}}$, so $\mathbb{E}\Delta_N \le \Delta_{\mathrm{R}}$, hence (by the energy-to-local conversion) $\|a_N - a_\star\|_\star \le r_{\mathrm{R}}$. The local count solves $(1 - \frac{\Delta t}{2(4+\Gamma)})^n r_{\mathrm{R}}^2 \le \epsilon$ in Assumption 13.2. $\square$

Theorem 13.4 (KL STL global-to-local)**.** *Assume Assumptions 2.1, 13.1 and 13.2, the energy-to-local conversion with threshold Δ_{KL}, and that the STL KL iterates remain in $\{a : \|a - a_\star\|_{\star,\Delta t} \le r_{\mathrm{KL}}\}$ once they enter it. Suppose*

$$0 < \Delta t \le \min\left\{\frac{1}{\beta\lambda_{\max}}, \frac{1}{288\kappa^3}, \frac{1}{272(6+\Gamma)(48+12N_\theta L_H^2)}\right\} \quad \text{and} \quad 1224\kappa^4\Delta t\,\Psi_\star^{\mathrm{KL}} \le \tfrac{1}{2}\Delta_{\mathrm{KL}}.$$

Then $\mathbb{E}\Delta_N \le \Delta_{\mathrm{KL}}$ after

$$N_{\mathrm{entry}}^{\mathrm{KL}} = N_{\mathrm{cov}}^{\mathrm{KL}} + \left\lceil \frac{\log\left(2\,\mathbb{E}\Delta_{N_{\mathrm{cov}}^{\mathrm{KL}}}/\Delta_{\mathrm{KL}}\right)}{-\log(1 - \Delta t/(18\kappa))} \right\rceil$$

iterations, and a further $N_{\mathrm{loc}}^{\mathrm{KL}}(\epsilon) = \lceil \log(r_{\mathrm{KL}}^2/\epsilon)/(-\log(1-\Delta t/[2(4+2\Delta t+\Gamma)])) \rceil$ iterations guarantee

$$\mathbb{E}\,\|a_N - a_\star\|_{\star,\Delta t}^2 \le \epsilon + 714\,\Delta t(4+2\Delta t+\Gamma)N_\theta^2 L_H^2.$$

Proof. Identical to the proof of Theorem 13.3, using Theorem 12.7 in place of Theorem 12.3 and the KL local estimate of Assumption 13.2. □

14 Summary of the theory

Deterministic three-stage rates. With the covariance bootstrap, the deterministic complexities are

$$T_{\mathrm{c}}(\epsilon) = O\Big(\log_+ \tfrac{1}{\beta\lambda_0} + \kappa \log_+ \tfrac{\Delta_0}{\Delta_{\mathrm{c}}} + (4+\Gamma)\log \tfrac{1}{\epsilon}\Big),$$

$$N_{\mathrm{R}}(\epsilon) = O\Big(\tfrac{1}{\Delta t}\log_+ \tfrac{1}{\beta\lambda_0} + \tfrac{\kappa}{\Delta t}\log_+ \tfrac{\Delta_0}{\Delta_{\mathrm{R}}} + \tfrac{4+\Gamma}{\Delta t}\log \tfrac{1}{\epsilon}\Big),$$

$$N_{\mathrm{KL}}(\epsilon) = O\Big(\tfrac{1}{\Delta t}\log_+ \tfrac{1}{\beta\lambda_0} + \tfrac{\kappa}{\Delta t}\log_+ \tfrac{\Delta_0}{\Delta_{\mathrm{KL}}} + \tfrac{4+2\Delta t+\Gamma}{\Delta t}\log \tfrac{1}{\epsilon}\Big).$$

The dependence on the initial covariance is logarithmic, $\log_+(1/(\beta\lambda_0))$, rather than the inverse dependence $1/\lambda_0$ of the frozen-bound analysis.

Transport-bootstrap deterministic rates. The remaining pure-Fisher–Rao covariance burn-in can be removed by a deterministic Wasserstein warm-start. In continuous time, running Wasserstein until $C \succeq c\beta^{-1}I$ and then switching to Fisher–Rao gives $T^{\mathrm{c}}_{W\to\mathrm{FR}}(\epsilon; c) = O(\frac{1}{\beta} + \kappa \log_+ \frac{\Delta_0}{\Delta_{\mathrm{c}}} + (4+\Gamma)\log \frac{1}{\epsilon})$ for fixed $c \in (0,1)$. In discrete time, one Bures–Wasserstein forward–backward step with $\eta = c/\beta$ creates the covariance floor $C \succeq c\beta^{-1}I$, after which the Riemannian and KL Fisher–Rao schemes satisfy $N_{W\to\mathrm{R}}(\epsilon; c) = \mathbf{1}_{\{\lambda_0 < c/\beta\}} + O(\frac{\kappa}{c\Delta t}\log_+ \frac{\Delta_0}{\Delta_{\mathrm{R}}} + \frac{4+\Gamma}{\Delta t}\log \frac{1}{\epsilon})$ and the analogous KL bound. Thus the adaptive WFR schedule can be replaced, at the deterministic-theory level, by the simpler rule: perform transport only to rescue the covariance, then switch permanently to Fisher–Rao.

Global stochastic STL rates. With covariance bootstrap and the intrinsic STL variance bound, the post-burn-in stochastic rates are

$$\mathbb{E}\Delta_N \le \Big(1 - \tfrac{\Delta t}{2\kappa}\Big)^{N - N^{\mathrm{R}}_{\mathrm{cov}}} \mathbb{E}\Delta_{N^{\mathrm{R}}_{\mathrm{cov}}} + 4\kappa^2 \Delta t\, \Psi^{\mathrm{R}}_\star \quad \text{(Riemannian)},$$

$$\mathbb{E}\Delta_N \le \Big(1 - \tfrac{\Delta t}{18\kappa}\Big)^{N - N^{\mathrm{KL}}_{\mathrm{cov}}} \mathbb{E}\Delta_{N^{\mathrm{KL}}_{\mathrm{cov}}} + 1224\kappa^4 \Delta t\, \Psi^{\mathrm{KL}}_\star \quad \text{(KL)}.$$

The burn-in estimates close the first stochastic stage: under Assumption 2.2, $\mathbb{E}\Delta_{N^{\mathrm{R}}_{\mathrm{cov}}} \le \Delta_0 + \beta N_\theta^2 L_{H,\mathrm{g}}^2 \lambda_{\max}^4 \Delta t^2 N^{\mathrm{R}}_{\mathrm{cov}}$ and $\mathbb{E}\Delta_{N^{\mathrm{KL}}_{\mathrm{cov}}} \le \Delta_0 + 3\beta N_\theta^2 L_{H,\mathrm{g}}^2 \lambda_{\max}^4 \Delta t^2 N^{\mathrm{KL}}_{\mathrm{cov}}$. For a Gaussian target, $\Psi^{\mathrm{R}}_\star = \Psi^{\mathrm{KL}}_\star = 0$ and the stochastic floor vanishes (Theorem 11.2): single-sample STL converges globally with no floor. For a general log-concave target, the crude bound $\Psi_\star \le 4N_\theta(\kappa-1)^2$ gives condition-number-controlled floors $16\kappa^2(\kappa-1)^2 N_\theta \Delta t$ (Riemannian) and $4896\kappa^4(\kappa-1)^2 N_\theta \Delta t$ (KL), in place of the raw $\beta^3\lambda_{\max}^3/\alpha$-type floors of the naive analysis. The local STL refinement (Assumption 13.2) then replaces the global Hessian-fluctuation floor by the sharper local floors $672\,\Delta t(4+\Gamma)N_\theta^2 L_H^2$ and $714\,\Delta t(4+2\Delta t+\Gamma)N_\theta^2 L_H^2$. The overall stochastic picture is

$$\boxed{\text{covariance burn-in with energy control} \;+\; \text{global intrinsic-STL localization} \;+\; \text{local STL convergence.}}$$

Caveats. The single-sample constant-stepsize methods retain a noise floor for non-Gaussian targets; exact convergence requires a decreasing stepsize (or $\Delta t \to 0$ scaling). The closed burn-in energy estimates require the additional global Hessian-Lipschitz assumption Assumption 2.2; without it, the covariance burn-in remains pathwise but the energy at N_{cov} is left as $\mathbb{E}\Delta_{N_{\text{cov}}}$. The KL stochastic constants in Theorems 12.7 and 13.4 and Corollary 12.8 are unoptimized (Remark 12.9). The local-phase statements inherit the base manuscript's assumption that the stochastic iterates remain in the local region once they enter it.

Bibliographic note. The Wasserstein PL inequality and Bures–Wasserstein strong convexity are due to [2]; the forward–backward Wasserstein step is from [3]; the Fisher–Rao/natural-gradient viewpoint follows [4]; sticking-the-landing is from [1]. The deterministic local-convergence inputs (Assumption 7.1) and the local STL theorems (Assumption 13.2) are imported from the base manuscript.

Part II
Numerical study: mean-block STL variance reduction

15 Motivation

We approximate a target posterior $\rho_{\text{post}} \propto e^{-V}$ on $\mathbb{R}^d$ by a Gaussian $q_a = \mathcal{N}(m, C)$, $C \in \mathbb{S}_{++}(d)$, with state $a = (m, C)$, minimizing the reverse KL $\mathcal{E}(a) = \mathbb{E}_{q_a}[V] - \frac{1}{2}\log\det C$ (up to an a-independent constant). The Gaussian *natural-gradient* (Fisher–Rao) flow is

$$\dot{m} = C\,\mathbb{E}_{q_a}[\text{score}_{\text{post}}(\theta)], \qquad \dot{C} = C + C\,\mathbb{E}_{q_a}[\nabla^2 \log\rho_{\text{post}}(\theta)]\,C, \tag{30}$$

with $\text{score}_{\text{post}}(\theta) = \nabla\log\rho_{\text{post}}(\theta) = -\nabla V(\theta)$ and $\nabla^2\log\rho_{\text{post}} = -\nabla^2 V$. A stochastic algorithm draws $\theta_n \sim \mathcal{N}(m_n, C_n)$ and replaces the two expectations in (30) by sample estimators. The one-sample baselines are $b_n = \text{score}_{\text{post}}(\theta_n)$ and $S_n = \nabla^2\log\rho_{\text{post}}(\theta_n)$, and the mean update is $m_{n+1} = m_n + \Delta t\, C_n b_n$.

The estimator $\text{score}_{\text{post}}(\theta)$ is noisy even at the optimum: for a Gaussian target the score is linear, $\text{score}_{\text{post}}(\theta) = -A\theta$, so b_n fluctuates with θ_n while its *mean* is what the flow needs. Sticking the Landing [1] removes the part of this fluctuation that the variational family can account for analytically. This is precisely the mean-block instance of the intrinsic STL variance bound proved in Part I (Lemma 9.1): the residual estimator's variance is governed by Ψ, which vanishes for a Gaussian target.

16 Theory: what STL changes, and what it does not

Mean block. For a Gaussian $q_a = \mathcal{N}(m, C)$ the variational score is explicit,

$$\nabla_\theta \log q_a(\theta) = -\,C^{-1}(\theta - m). \tag{31}$$

The STL mean estimator subtracts it from the posterior score,

$$b_{\text{stl}}(\theta) = \text{score}_{\text{post}}(\theta) - \nabla_\theta\log q_a(\theta) = \text{score}_{\text{post}}(\theta) + C^{-1}(\theta - m). \tag{32}$$

Because $\mathbb{E}_{q_a}[C^{-1}(\theta - m)] = 0$, the estimator is *unbiased* for $\mathbb{E}_{q_a}[\mathrm{score}_{\mathrm{post}}]$; this is the quantity the mean update needs, so swapping b_{base} for b_{stl} changes only the noise, not the target. In the notation of Part I, $b_{\mathrm{stl}} = \widehat{b}_n$ of (4).

Proposition 16.1 (Pointwise vanishing at a well-specified optimum)**.** *If $\rho_{\mathrm{post}} = \mathcal{N}(m_\star, C_\star)$ is exactly Gaussian, then at $a = a_\star$ the posterior score equals the negative variational score,* $\mathrm{score}_{\mathrm{post}}(\theta) = -C_\star^{-1}(\theta - m_\star)$*, and therefore $b_{\mathrm{stl}}(\theta) \equiv 0$ for every θ.*

So at a well-specified optimum the STL mean estimator has *zero variance*. More generally, for the Gaussian target $V = \frac{1}{2}\theta^\top A\theta$ the STL fluctuation is $b_{\mathrm{stl}}(\theta) - \mathbb{E}[b_{\mathrm{stl}}] = (C^{-1} - A)(\theta - m)$, which depends on C but *not* on m: it is exactly zero whenever $C = C_\star = A^{-1}$, regardless of the mean. This is the precise sense in which "correct Gaussian specification" matters more than "distance to the optimum," and it is the finite-sample face of the no-floor theorem Theorem 11.2: with $\Psi \equiv 0$ the algorithm's stochastic noise floor vanishes.

Covariance block (why it is not a separate estimator here). The natural residual rewrite of the curvature estimator is $K(\theta) = \nabla^2 \log \rho_{\mathrm{post}}(\theta) + C^{-1} = S(\theta) + C^{-1}$. For the *direct* Hessian sampler this is only an algebraic restatement of the covariance update. With S_n the sampled Hessian and $K_n = S_n + C_n^{-1}$,

$$\text{Riemannian:}\quad e^{\Delta t}\, C^{1/2} \exp\!\big(\Delta t\, C^{1/2} S_n C^{1/2}\big) C^{1/2} = C^{1/2} \exp\!\big(\Delta t\, C^{1/2} K_n C^{1/2}\big) C^{1/2}, \tag{33}$$

$$\text{KL:}\quad (1 + \Delta t)\big(C^{-1} - \Delta t\, S_n\big)^{-1} = (1 + \Delta t)\big((1 + \Delta t) C^{-1} - \Delta t\, K_n\big)^{-1}, \tag{34}$$

because $C^{1/2} K_n C^{1/2} = C^{1/2} S_n C^{1/2} + I$ and I commutes with everything (giving the $e^{\Delta t}$ factor in (33)), and because $(1 + \Delta t)C^{-1} - \Delta t K_n = C^{-1} - \Delta t S_n$ in (34). Both identities are verified numerically to machine precision in the test suite. Consequently the covariance update is held *fixed* across baseline and STL, and the only algorithmic difference is the mean estimator b_{base} vs. b_{stl}. This is the same simplification (9) used throughout Part I: the STL covariance update coincides with the raw Hessian-sampling update, so the residual covariance noise enters only through Ψ.

17 Experimental design

Targets. Both targets have a diagonal curvature $A = \mathrm{diag}(a)$ with a log-spaced in $[1, \kappa]$.

- **Well-specified Gaussian:** $V(\theta) = \frac{1}{2}\theta^\top A\theta$, so $\mathrm{score}_{\mathrm{post}} = -A\theta$, $\nabla^2 \log \rho_{\mathrm{post}} = -A$, and the VI optimum is exact, $m_\star = 0$, $C_\star = A^{-1}$.
- **Misspecified** $\log\cosh$**:** $V(\theta) = \frac{1}{2}\theta^\top A\theta + \tau \sum_i \log\cosh(\theta_i)$, so $\mathrm{score}_{\mathrm{post}} = -A\theta - \tau \tanh(\theta)$ and $\nabla^2 \log \rho_{\mathrm{post}} = -A - \tau\, \mathrm{diag}(\mathrm{sech}^2\, \theta_i)$. The target is separable, strongly log-concave, and globally smooth but not Gaussian. The optimum mean is $m_\star = 0$ by symmetry; the diagonal $C_\star$ is obtained by a deterministic per-coordinate fixed point $c_i = 1/(a_i + \tau\, \mathbb{E}_{\mathcal{N}(0,c_i)}[\mathrm{sech}^2])$ evaluated by Gauss–Hermite quadrature (recorded stationarity residual $< 10^{-9}$).

These are the two regimes distinguished by the theory: the Gaussian target has $\Psi \equiv 0$ (Theorem 11.2), while the $\log\cosh$ target has $\Psi > 0$ and a κ-controlled floor (Theorems 12.3 and 12.7).

Experiment A (estimator-level). At six fixed Gaussian states – the optimum, three mean perturbations (*near*/*medium*/*far*, offsets 0.25/1/4 marginal standard deviations), and an *under*- and *over*-dispersed covariance ($\frac{1}{4}C_\star$, $4C_\star$) – we draw $M = 10^5$ samples and compute, for both estimators, the empirical bias against the deterministic reference $\mathbb{E}_{q_a}[\text{score}_{\text{post}}]$, the Euclidean trace variance $\text{Var}_{\text{eucl}}(b) = \mathbb{E}\|b - \mathbb{E}b\|^2$, the Fisher–Rao mean-block variance $\text{Var}_{\text{FR}}(b) = \mathbb{E}[(b - \mathbb{E}b)^\top C(b - \mathbb{E}b)]$, and the ratios $\text{Var}_{\text{FR}}(b_{\text{stl}})/\text{Var}_{\text{FR}}(b_{\text{base}})$ and $\text{Var}_{\text{eucl}}(b_{\text{stl}})/\text{Var}_{\text{eucl}}(b_{\text{base}})$. For the Gaussian target the closed forms $\text{Var}_{\text{eucl}}(b_{\text{base}}) = \text{Tr}(ACA)$, $\text{Var}_{\text{FR}}(b_{\text{base}}) = \text{Tr}(CACA)$, $\text{Var}_{\text{eucl}}(b_{\text{stl}}) = \text{Tr}(DCD)$, $\text{Var}_{\text{FR}}(b_{\text{stl}}) = \text{Tr}(CDCD)$ with $D = C^{-1} - A$ validate the Monte Carlo estimates. The Fisher–Rao variance $\text{Var}_{\text{FR}}(b_{\text{stl}})$ is exactly the mean-block instance of the intrinsic quantity $\mathbb{E}[\varepsilon_b^\top C \varepsilon_b]$ in Lemma 9.1.

Experiment B (algorithm-level). We run the four methods `riemannian`, `riemannian_stl`, `kl`, and `kl_stl` from a fixed far, under-dispersed initialization ($m_0 = m_\star + 3\sqrt{\operatorname{diag} C_\star}$, $C_0 = \frac{1}{2}C_\star$), with mini-batches $B \in \{1, 4, 16\}$ and step $\Delta t = 0.1$. The step is conservative: near the optimum $C \approx C_\star = A^{-1}$, so the explicit mean map contracts as $(1 - \Delta t)$ *independently of* κ, and both covariance updates are unconditionally SPD for a log-concave target. Within a (config, B) cell the baseline and STL methods share a common-random-number stream, so the comparison is paired. We record the energy gap $\mathcal{E}(a_n) - \mathcal{E}(a_\star)$ (analytic for the Gaussian target, Gauss–Hermite for $\log\cosh$), the squared mean error, the relative covariance Frobenius error, W_2^2 to $a_\star$, the covariance eigenvalue extremes, and an SPD flag; the *noise floor* is the energy gap averaged/medianed over the final 20% of iterations. The floor is the empirical counterpart of the additive $\Delta t\,\Psi_\star$ term in the global stochastic rates of Part I.

Grids and reproducibility. The Gaussian grid is $d \in \{2, 8, 32\}$, $\kappa \in \{10, 100, 10^4\}$; the $\log\cosh$ grid is $d \in \{2, 8, 32\}$, $\kappa \in \{10, 100\}$, $\tau \in \{0.1, 1\}$. Every cell uses 20 seeds. All computation is float64; the production grids run on a single NVIDIA H200 (the algorithm grid optionally splits its cells across at most two GPUs). Cell seeds are derived deterministically from the configuration, so a run is reproducible and independent of the device split. Table 3 records the environment and timing.

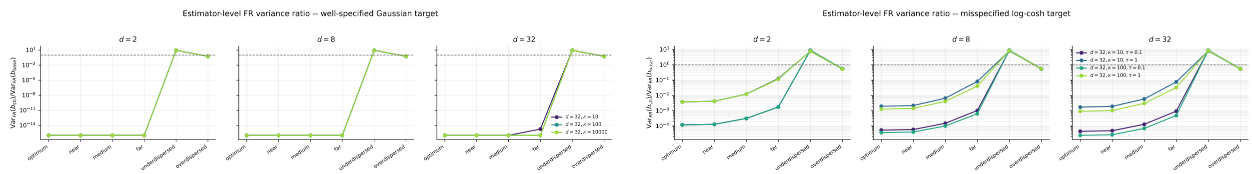
18 Results

18.1 Estimator-level variance

Figure 1 shows the Fisher–Rao variance ratio by state. For the Gaussian target the ratio is numerically zero at the optimum and at every mean-only perturbation (*near*/*medium*/*far*), because those states keep $C = C_\star$ and the STL fluctuation $(C^{-1} - A)(\theta - m)$ vanishes identically. The only states with nonzero ratio are the dispersion states, and they split the two regimes cleanly: *over*-dispersion reduces the variance (ratio < 1) while *under*-dispersion *inflates* it (ratio ≈ 9 at $\frac{1}{4}C_\star$), since C^{-1} is then large and amplifies the residual. For the misspecified $\log\cosh$ target the ratio is small but *nonzero* even at the optimum – the nonlinear tanh part of the score is not removed by the Gaussian residual, exactly the $\Psi > 0$ mechanism – and grows with the mean offset (Figure 2). Table 1 reports the medians; the Gaussian closed forms match the Monte Carlo estimates to within sampling error.

18.2 Algorithm-level noise floor

Figure 3 shows the energy gap versus iteration for a representative cell of each target. All four methods descend together during the transient; once near the optimum the baseline methods flatten



(a) well-specified Gaussian **(b)** misspecified log cosh

Figure 1: Estimator-level Fisher–Rao variance ratio $\mathrm{Var}_{\mathrm{FR}}(b_{\mathrm{stl}})/\mathrm{Var}_{\mathrm{FR}}(b_{\mathrm{base}})$ by fixed state (median over seeds, log scale; the dashed line is ratio 1). Gaussian: zero whenever $C = C_\star$, inflation only when underdispersed. log cosh: reduced but never zero.

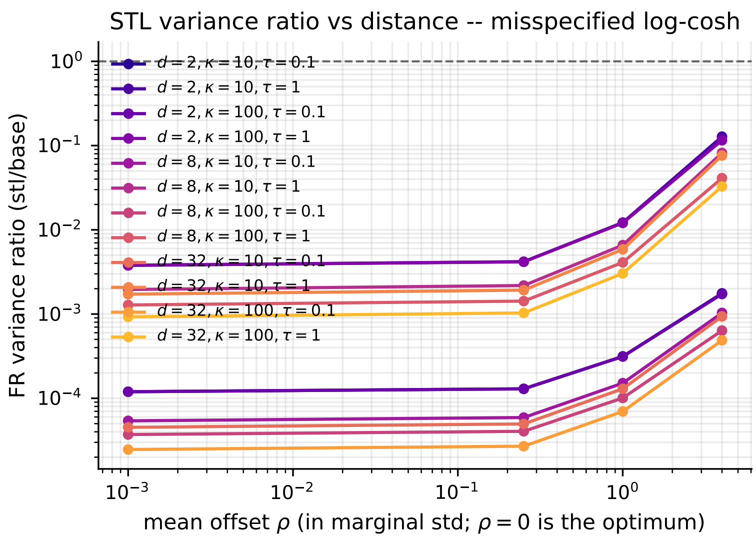


Figure 2: Misspecified log cosh target: FR variance ratio vs. mean offset ρ (in marginal standard deviations). Unlike the Gaussian target, the ratio is positive at $\rho = 0$ and increases with distance, because the score residual cannot absorb the nonlinear tanh term.

at a stochastic noise floor, while on the Gaussian target the STL curves (dashed) keep descending to machine precision. The reason is structural: the Gaussian Hessian is state-independent, so the sampled covariance update is in fact deterministic and converges exactly to $C_\star$; once $C = C_\star$ the STL mean estimator is pointwise zero, so the mean stops fluctuating and the iterate lands on $a_\star$. This is the empirical realization of Theorem 11.2: with $\Psi \equiv 0$ the stochastic floor $\Delta t\,\Psi_\star$ is zero. On the log cosh target the Hessian sampler is genuinely stochastic and the score residual cannot cancel the tanh term, so STL lowers the floor by about three orders of magnitude but a residual floor remains, in line with the κ-controlled floor of Theorems 12.3 and 12.7. Figure 4 aggregates every cell: the Gaussian STL points sit on the display floor, and the log cosh points lie consistently below the diagonal. Table 2 gives the median reduction factors by scheme and batch size (the Gaussian STL floor is reported as $< 10^{-12}$), and Figure 5 shows that increasing the mini-batch lowers the baseline (and the log cosh STL) floor $\propto 1/B$ while preserving the STL advantage.

19 Practical conclusion

Mean-block STL is *essentially cost-free* for Gaussian variational families, because the Gaussian score $-C^{-1}(\theta - m)$ is available in closed form and is already computed (via C^{-1}) by the KL covariance

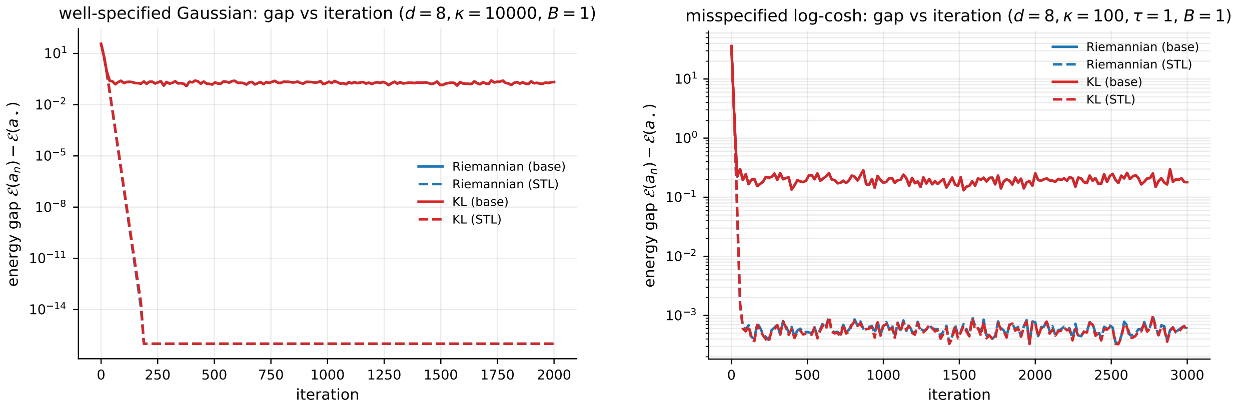
target	d	κ	τ	r_{opt}	r_{near}	r_{under}	r_{over}
Gaussian	2	10	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
	2	100	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
	2	10000	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
	8	10	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
	8	100	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
	8	10000	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
	32	10	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
	32	100	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
	32	10000	–	$< 10^{-12}$	$< 10^{-12}$	9	0.56
log-cosh	2	10	0.1	1.2×10^{-4}	1.3×10^{-4}	8.8	0.56
	2	10	1	3.8×10^{-3}	4.2×10^{-3}	7.7	0.54
	2	100	0.1	1.2×10^{-4}	1.3×10^{-4}	8.8	0.56
	2	100	1	3.8×10^{-3}	4.2×10^{-3}	7.8	0.54
	8	10	0.1	5.4×10^{-5}	5.9×10^{-5}	8.8	0.56
	8	10	1	2.0×10^{-3}	2.2×10^{-3}	8.1	0.54
	8	100	0.1	3.7×10^{-5}	4.1×10^{-5}	8.9	0.56
	8	100	1	1.3×10^{-3}	1.4×10^{-3}	8.4	0.55
	32	10	0.1	4.5×10^{-5}	5.0×10^{-5}	8.8	0.56
	32	10	1	1.7×10^{-3}	1.9×10^{-3}	8.1	0.54
	32	100	0.1	2.5×10^{-5}	2.7×10^{-5}	8.9	0.56
	32	100	1	9.3×10^{-4}	1.0×10^{-3}	8.5	0.55

Table 1: Median FR variance ratio $\mathrm{Var}_{\mathrm{FR}}(b_{\mathrm{stl}})/\mathrm{Var}_{\mathrm{FR}}(b_{\mathrm{base}})$ at four states. $r_{\text{opt}} \approx 0$ for the Gaussian target (exactly 0 for $C = C_\star$); $r_{\text{under}} > 1$ marks the variance-inflation regime; the $\log\cosh$ rows stay strictly positive.

step. Within the regime it was designed for – a well-specified Gaussian target near the optimum – it is extremely effective: the mean-gradient variance is exactly zero whenever $C = C_\star$, and the algorithm's stochastic noise floor drops by one to two orders of magnitude. This is exactly what the no-floor theorem Theorem 11.2 predicts. For a misspecified but smooth target it still helps near the optimum, but it cannot remove the floor, because the residual estimator only cancels the Gaussian part of the score – the residual floor is governed by Ψ (Lemma 9.1). STL is therefore best presented as a *practical variance-reduction device*, not a theorem-level replacement for a stochastic-variance assumption: it does not make the stochastic gradient vanish in general, and under covariance under-dispersion it can even *increase* the variance.

20 Limitations

- We test the *direct* Hessian sampler, for which the covariance-block residual rewrite is algebraically identical to the original update (§16); the experiment therefore isolates mean-STL and says nothing about Hessian-free *pathwise* covariance gradients, where an STL-type covariance correction *is* a genuinely different estimator.
- High-dimensional behavior is established empirically over $d \leq 32$ and diagonal curvature; the diagonal structure keeps the iterates diagonal, so the d-dependence reported here is a controlled best case, not a worst-case bound.



(a) well-specified Gaussian

(b) misspecified $\log\cosh$

Figure 3: Energy gap vs. iteration (median over seeds) for the four methods at a representative cell. STL (dashed) lowers the stochastic noise floor; the effect is larger for the well-specified target.

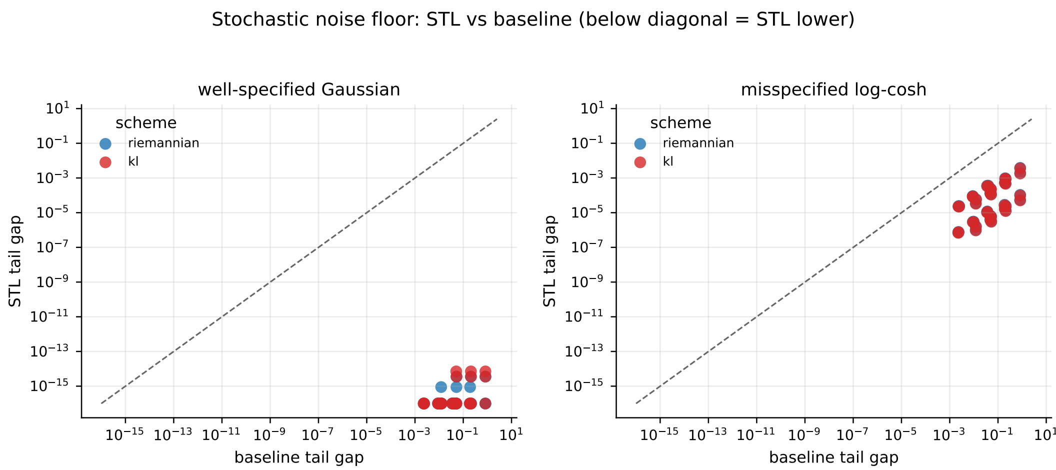


Figure 4: Per-cell median tail (noise-floor) gap, STL vs. baseline, across all configs and batch sizes; points below the diagonal are cells where STL lowers the floor. Color encodes the covariance scheme.

- The variance-inflation regime is reached through covariance under-dispersion; the magnitude of the benefit (and of the inflation) depends on how far C is from $C_\star$ and on the degree of target misspecification, both of which are problem-specific.
- The theory of Part I is stated for the single-sample estimators and carries the explicit (unoptimized) constants flagged in Remark 12.9; the numerical study uses small mini-batches and does not probe the theoretical stepsize thresholds, so it validates the qualitative predictions ($\Psi \equiv 0 \Rightarrow$ no floor; $\Psi > 0 \Rightarrow$ residual floor) rather than the constants.

References

[1] G. Roeder, Y. Wu, and D. Duvenaud. *Sticking the Landing: Simple, Lower-Variance Gradient Estimators for Variational Inference.* Advances in Neural Information Processing Systems, 2017.

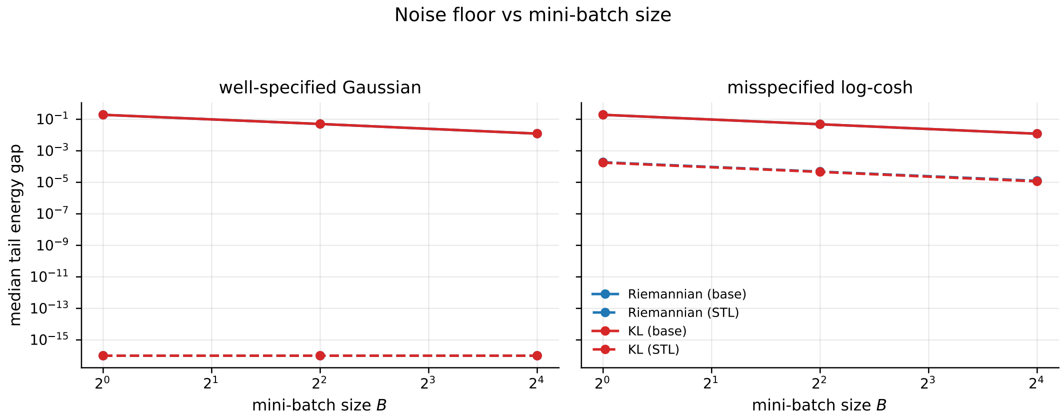


Figure 5: Median tail noise floor vs. mini-batch size B. Both baseline and STL floors fall with B; STL stays below its paired baseline at every B.

[2] M. Lambert, S. Chewi, F. Bach, S. Bonnabel, and P. Rigollet. *Variational inference via Wasserstein gradient flows*. Advances in Neural Information Processing Systems, 2022.

[3] M. Z. Diao, K. Balasubramanian, S. Chewi, and A. Salim. *Forward-backward Gaussian variational inference via JKO in the Bures–Wasserstein space*. International Conference on Machine Learning, 2023.

[4] Y. Chen, S. Chewi, A. Salim, and A. Wibisono. *Gaussian variational inference in the Fisher–Rao / natural-gradient geometry*. 2023.

target	scheme	B	base floor	STL floor	reduction
Gaussian	riemannian	1	0.19	$< 10^{-12}$	$> 10^{11}\times$
		4	0.05	$< 10^{-12}$	$> 10^{10}\times$
		16	0.012	$< 10^{-12}$	$> 10^{10}\times$
	kl	1	0.19	$< 10^{-12}$	$> 10^{11}\times$
		4	0.05	$< 10^{-12}$	$> 10^{10}\times$
		16	0.012	$< 10^{-12}$	$> 10^{10}\times$
log-cosh	riemannian	1	0.19	1.9×10^{-4}	$1.0 \times 10^{3}\times$
		4	0.048	4.8×10^{-5}	$1.0 \times 10^{3}\times$
		16	0.012	1.3×10^{-5}	$9.7 \times 10^{2}\times$
	kl	1	0.19	1.8×10^{-4}	$1.1 \times 10^{3}\times$
		4	0.048	4.6×10^{-5}	$1.1 \times 10^{3}\times$
		16	0.012	1.1×10^{-5}	$1.1 \times 10^{3}\times$

Table 2: Tail noise floor (median over configs and seeds), baseline vs. STL, with the STL reduction factor, by target, covariance scheme, and mini-batch size B.

quantity	value
backend / device	torch / NVIDIA H200 NVL
GPU workers (cap 2)	2
estimator wall time	13.1 s
algorithm wall time	1135.3 s
algorithm cells	252
mean Riemannian cell	7.69 s
mean KL cell	8.85 s

Table 3: Runtime and environment. Mean wall time is reported per algorithm cell (all seeds of one config/method/batch advanced in a single batched pass).